Classical and Bayesian estimation of multicomponent stress-strength reliability for the Weibull distribution

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Abstract

We study a system that works when at least three of five component strengths exceed a common random stress. Strength and stress follow Weibull distributions with the same shape and different rates. Strength observations are collected in blocks under progressive hybrid censoring, and stress observations are upper k-record values. We derive the reliability function and estimate it by unrestricted and order-restricted maximum likelihood. We also obtain Bayesian estimates under a dependent gamma-Dirichlet prior. A Monte Carlo study with 48 scenarios and 300 replications per scenario compares absolute bias and mean squared error. Asymptotic, equal-tail and numerical highest posterior density intervals are compared through coverage and average width. An incorrect order restriction increases bias and can prevent intervals from covering the true reliability. Published jute-fibre and carbon-fibre measurements illustrate the methods. The jute example has a Weibull-fit limitation; the additional carbon example is compatible with the fitted model.

Keywords: Weibull distribution; stress-strength reliability; progressive hybrid censoring; upper k-records; maximum likelihood; gamma-Dirichlet prior.

1 Introduction

Reliability is the probability that a system performs its intended task. In a stress-strength model, a component works when its strength is greater than the stress applied to it. A system with several components can continue to work even if some components fail. Here we consider an s-out-of-K system: at least s of the K strengths must exceed the same stress.

Complete observations are not always available in a life test. Some specimens may be withdrawn before failure and the experiment may end at a time limit. Progressive hybrid censoring describes these two features. A block represents a separate test facility. For the stress sample, we retain upper k-records instead of every observation. These records are dependent, so their likelihood differs from the likelihood of independent Weibull observations.

Sultana et al. [1] study block Type-I hybrid censored Weibull populations, including order-restricted inference and a dependent prior. Newer [2] combines block adaptive progressive hybrid censoring with upper k-record sampling for another distribution. Newer and Alanazi [3] discuss multicomponent reliability, Bayesian estimation and bootstrap inference under a different sampling design. We use these papers to develop the assigned Weibull model. We do not use their other distributions or ranked-set likelihoods in place of the model below.

The order assumption in this project is that the strength population is stochastically larger than the stress population. It can improve estimation when it is correct, but it can also produce misleading results when it is false. We therefore include correctly ordered, boundary and incorrectly ordered cases in the simulation. Sections 2 and 3 present the classical and Bayesian estimates. Section 4 contains the simulation and the data examples. Section 5 gives the conclusions.

1.1 What we did in this project

The main question is simple: how likely is a system to work when at least three of its five components must be stronger than the same applied stress? The observations available for estimation are incomplete. Some strength specimens leave a test before failure, and only selected record values of stress are retained.

We carried out the following steps.

1. We read the supplied papers and selected the Weibull distribution required by the assigned topic. We kept the strength and stress roles separate throughout the calculation.
2. We derived the shared-stress reliability probability, then wrote a likelihood for the actual strength censoring and stress record schemes.
3. We found maximum likelihood estimates with and without the rate-order restriction. The ordered calculation includes reoptimizing the common shape.
4. We specified a dependent joint prior and calculated posterior means, equal-tail intervals and numerical HPD intervals. The prior was fixed before the simulation study.
5. We generated 14,400 simulated datasets across 48 settings. Each dataset was given to all four estimators so that differences between methods did not come from using different random samples.
6. We transcribed two published datasets, checked distributional assumptions and imposed reproducible retrospective censoring and record extraction. The complete measurements and retained observations were saved separately.
7. We checked numerical calculations and examined changes caused by the prior, stopping scheme and acquisition order. The report includes both the useful results and the limitations.

The result is a comparison of estimation methods for one stated model. It is not a new experiment on manufactured specimens, and it does not establish a certified operating reliability for a real installation.

1.2 How to read the results

There are three different kinds of data in the project. The raw jute and carbon measurements are published observations. The censored blocks and records are retrospective selections made from those observations. The Monte Carlo datasets are generated under known parameter values to test the estimators. These should not be confused with each other.

The point-estimation comparison asks whether an estimate is close to the true value over repeated simulated datasets. The interval comparison asks how often an interval covers that value and how wide it is. A short interval is helpful only if its coverage is also reasonable. A credible interval describes posterior probability under a prior and a likelihood; a confidence interval has a repeated-sampling interpretation. We report both types but do not give them the same interpretation.

For the data examples, the population truth is unknown. We can report fitted estimates, intervals, goodness-of-fit checks and sensitivity results, but we cannot calculate an observed-data MSE against an unknown truth. Simulation supplies that part of the comparison.

1.3 Main notation

1.4 Weibull model and system reliability

Let $X_1, \ldots, X_K$ be independent strengths, and let Y be a stress independent of them. All strengths have rate a and stress has rate b, with common shape $\alpha > 0$. For a positive Weibull variable with

Table 1: Symbols used throughout the report. The record order k and system size K have different meanings.

Symbol	Meaning
α	Common Weibull shape parameter
a, b	Strength and stress rate parameters
K, s	Number of components and minimum number that must survive
k, r	Stress record order and prescribed number of records
n_i, m_i, T_i	Initial block size, target failures and time threshold
d_i, r_{ij}, q_i	Observed failures, actual withdrawals and terminal survivors
d, q	Total strength failures and $q = d + r$
$A(\alpha), B(\alpha)$	Strength and stress exposure terms in the likelihood
R	The 3-out-of-5 shared-stress reliability in the numerical work
M	Number of Monte Carlo replications within a scenario

rate $\lambda > 0$,

$$f(z;\alpha,\lambda) = \alpha\lambda z^{\alpha-1}e^{-\lambda z^{\alpha}}, \qquad z > 0, \tag{1}$$

$$F(z) = 1 - e^{-\lambda z^{\alpha}}, \qquad S(z) = e^{-\lambda z^{\alpha}}, \tag{2}$$

$$h(z) = \alpha\lambda z^{\alpha-1}, \qquad F^{-1}(u) = \left[\frac{-\log(1-u)}{\lambda}\right]^{1/\alpha}. \tag{3}$$

The usual scale is $\lambda^{-1/\alpha}$. We call λ a rate to make the order assumption clear: $a \le b$ means that strength has greater survival than stress at every positive threshold. The shapes below and above one give decreasing and increasing hazards, respectively.

Conditional on $Y = y$, the number of surviving components is binomial with probability $e^{-ay^{\alpha}}$. Thus

$$R_{s,K} = \int_0^{\infty} \sum_{j=s}^{K} \binom{K}{j} e^{-jay^{\alpha}} (1 - e^{-ay^{\alpha}})^{K-j} f_Y(y)\, dy. \tag{4}$$

Put $\rho = b/a$ and $u = e^{-ay^{\alpha}}$. Since $f_Y(y)dy = -\rho u^{\rho-1}du$, integration gives

$$R_{s,K} = \rho \sum_{j=s}^{K} \binom{K}{j} B(\rho + j, K - j + 1). \tag{5}$$

An equivalent expression is easier to evaluate. The transformed strengths X_i^{α} are exponential with rate a. The $(K - s + 1)$-th smallest transformed strength is a sum of independent exponentials with rates $Ka, (K-1)a, \ldots, sa$. Taking its Laplace transform at b gives

$$\boxed{R_{s,K} = 1 - \prod_{j=s}^{K} \frac{j}{j+\rho}.} \tag{6}$$

For $K = s = 1$ this reduces to $b/(a + b)$. Reliability tends to zero as $\rho \to 0$ and to one as $\rho \to \infty$. It does not depend directly on the common shape, although the fitted rates do. We use $s = 3$ and $K = 5$ throughout the numerical study. In this case $R_{3,5} = 1/2$ at $a = b$, and the order restriction implies $R_{3,5} \ge 1/2$.

The common stress in (4) matters. Component survival indicators are independent conditional on Y, but dependent after averaging over Y. Substituting $P(X > Y)$ directly into an ordinary binomial probability would describe a different experiment. The notebook checks (6) by numerical integration and direct simulation.

1.4.1 Distribution and reliability graphs

The four distribution plots use rate one and several shapes. With shape below one, the hazard decreases; with shape one it is constant; with shape above one it increases. These are properties of the chosen distribution, not fitted statements about the published specimens. A density may be greater than one at a point; its area, rather than its height, is a probability.

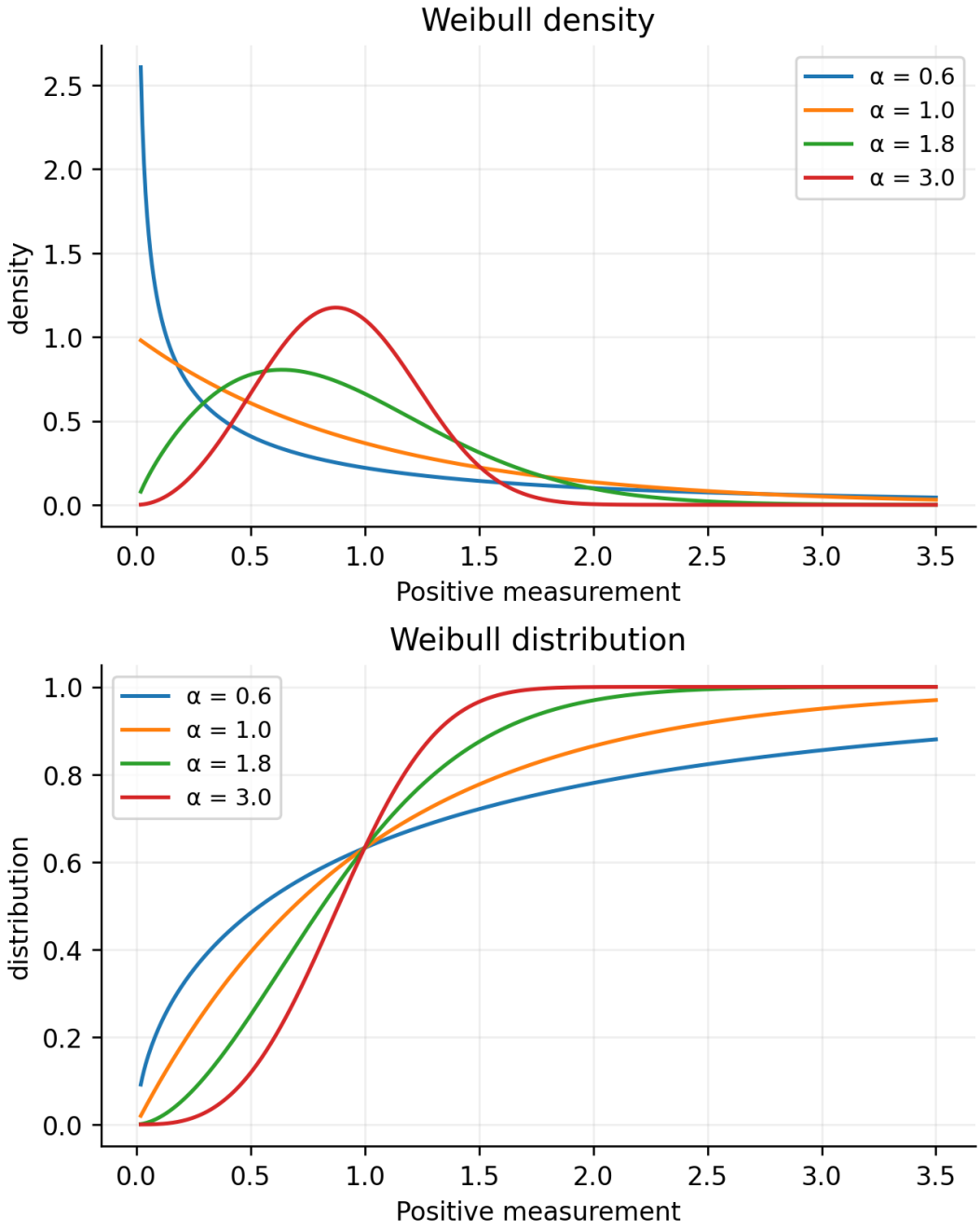


Figure 1: Weibull density and cumulative distribution for shapes 0.6, 1, 1.8 and 3 at rate one.

A larger stress rate makes large stress values less likely in this rate parameterization, so reliability increases with b/a. Requiring more components to survive reduces reliability. The second reliability panel compares the common-stress model with a model that gives every component an independent stress. The two curves differ because shared stress creates dependence between component survival indicators.

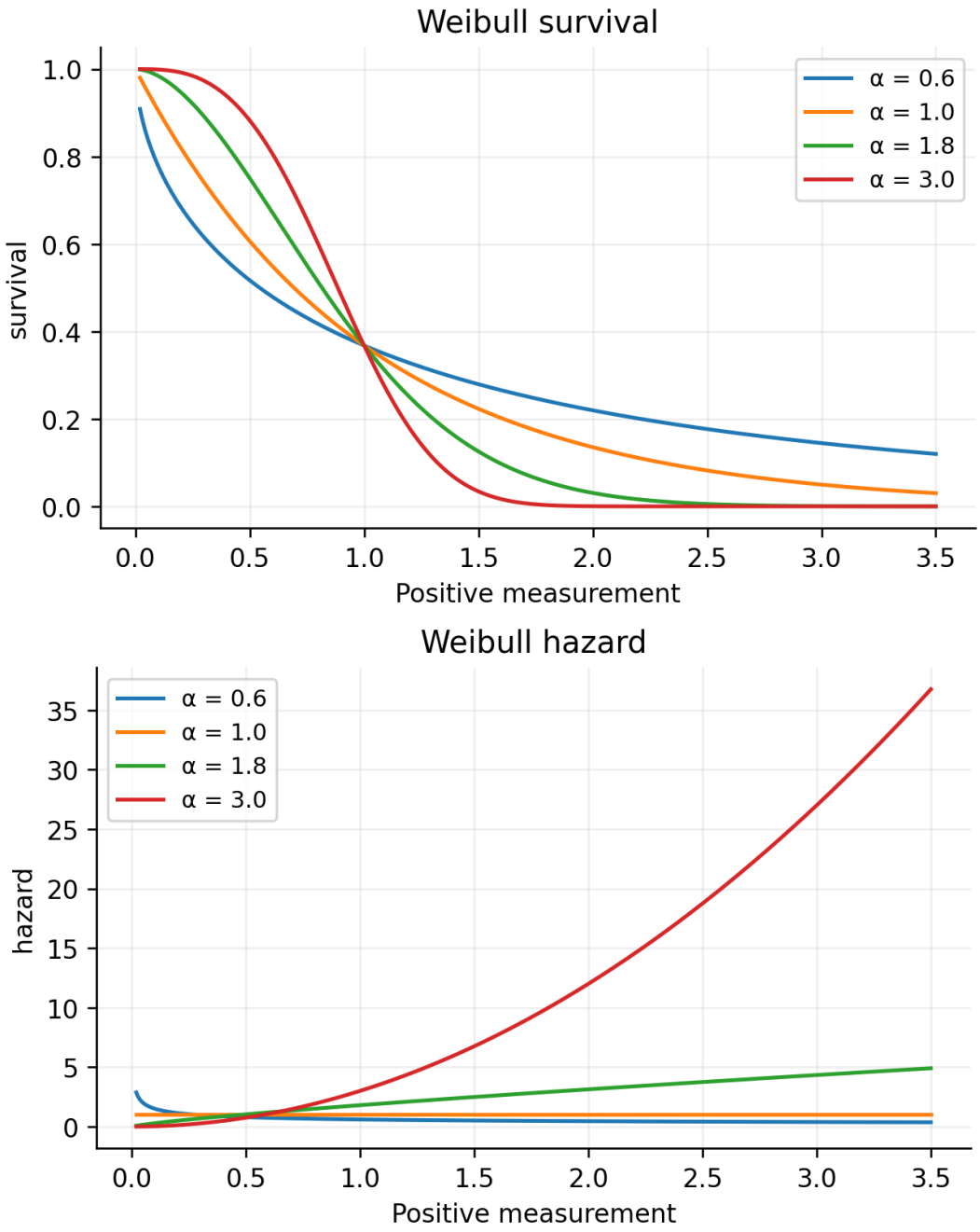


Figure 2: Weibull survival and hazard for the same parameter choices. Survival decreases with the threshold, while the shape controls the hazard trend.

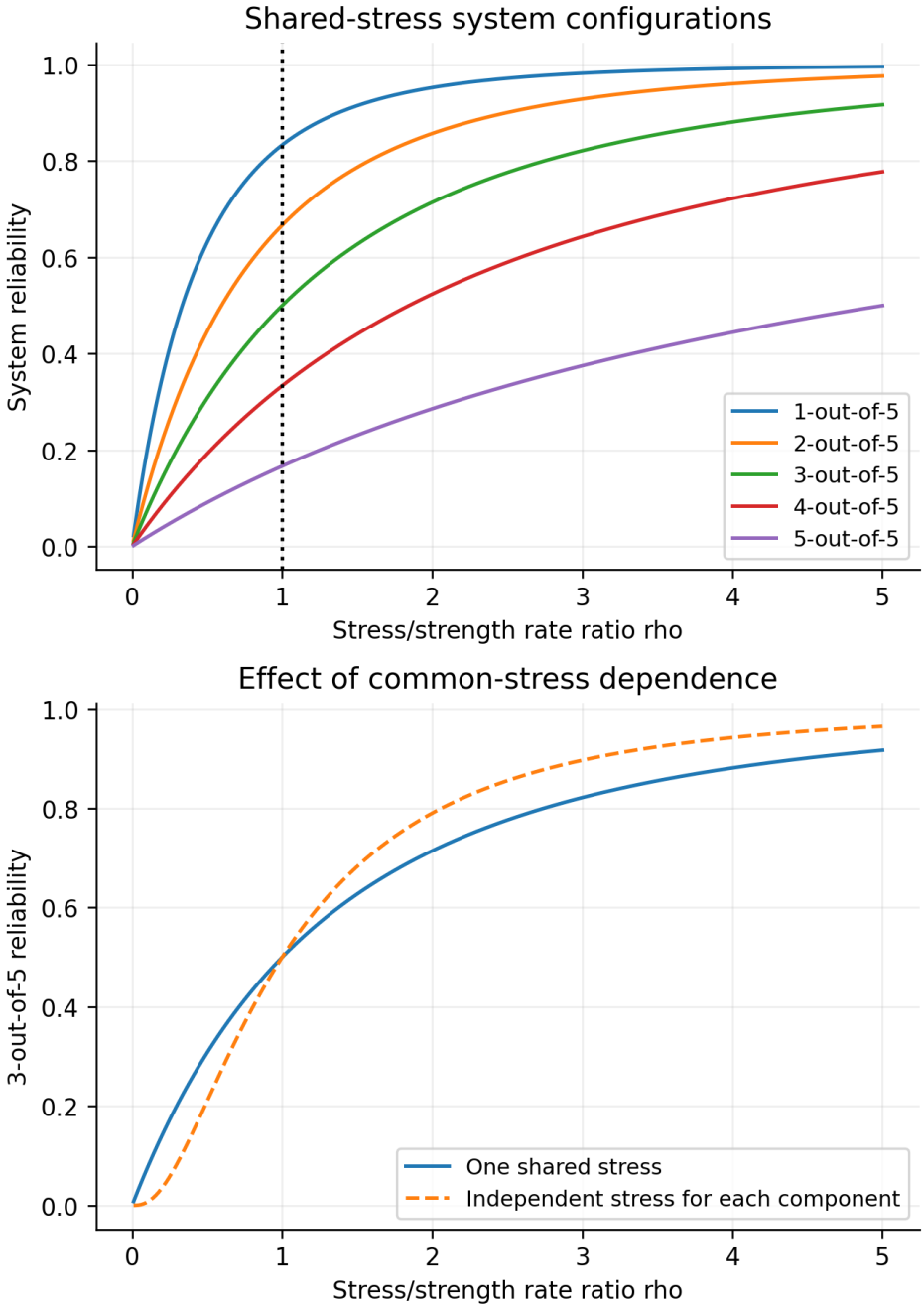


Figure 3: Reliability for several system requirements, and the effect of using one shared stress instead of separate independent stresses.

1.5 Observation schemes and likelihood

For strength block i, start with n_i specimens. Choose a target number of failures m_i, a time T_i, and planned withdrawals R_{ij} satisfying $\sum_j R_{ij} = n_i - m_i$. Withdraw survivors uniformly, independently of their unobserved failure times.

The principal Type-I scheme stops at $\tau_i = \min(x_{i,m_i}, T_i)$. Let d_i be the observed number of failures and r_{ij} the actual withdrawals. If the target is reached, all remaining survivors are counted among the final withdrawals. Otherwise the remaining $q_i = n_i - d_i - \sum_j r_{ij}$ specimens are censored at T_i. Up to parameter-free design factors,

$$L_i \propto \prod_{j=1}^{d_i} f_X(x_{ij}) S_X(x_{ij})^{r_{ij}} S_X(\tau_i)^{q_i}. \tag{7}$$

There is no double counting: $q_i = 0$ when final survivors have already been withdrawn. Blocks with no failures still contribute their survival factors. The adaptive Type-II alternative guarantees m_i failures. Withdrawals after T_i are cancelled until all survivors are removed at the target failure. The same likelihood applies using the actual withdrawals. This experiment can continue beyond T_i.

For stress, the first upper k-record is the minimum of the first k observations. Thereafter retain a new record whenever the k-th largest observed value increases. Stop after r distinct records $y_1 < \cdots < y_r$. On the cumulative-hazard scale, successive record increments are independent exponential variables with rate k. The joint record-value likelihood is therefore

$$L_Y = k^r S_Y(y_r)^k \prod_{j=1}^{r} h_Y(y_j) \propto \alpha^r b^r \exp\left[(\alpha - 1)\sum_j \log y_j - bky_r^\alpha\right]. \tag{8}$$

We use this marginal record-value experiment. Acquisition positions are saved to explain extraction, but their waiting times are not included in (8).

Write $d = \sum_i d_i$, $q = d + r$, and define

$$A(\alpha) = \sum_{i,j}(1 + r_{ij})x_{ij}^\alpha + \sum_i q_i \tau_i^\alpha, \tag{9}$$

$$B(\alpha) = ky_r^\alpha, \qquad C = \sum_{i,j} \log x_{ij} + \sum_j \log y_j. \tag{10}$$

The joint log-likelihood, apart from constants, is

$$\ell(\alpha, a, b) = q \log \alpha + d \log a + r \log b + (\alpha - 1)C - aA(\alpha) - bB(\alpha). \tag{11}$$

All calculations below use (11), with positive parameters. The numerical experiments have $d > 0$ and at least two distinct stress records.

2 Maximum Likelihood Estimation

2.1 MLE under no order restriction

For fixed α, differentiation of (11) with respect to the two rates gives

$$\widehat{a}(\alpha) = \frac{d}{A(\alpha)}, \qquad \widehat{b}(\alpha) = \frac{r}{B(\alpha)}. \tag{12}$$

Substitution gives the profile log-likelihood

$$\ell_p(\alpha) = q \log \alpha - d \log A(\alpha) - r \log B(\alpha) + (\alpha - 1)C + \text{constant}. \tag{13}$$

We maximize it over positive α and then use (12). The profile score is

$$\ell_p'(\alpha) = \frac{q}{\alpha} + C - d\frac{A'(\alpha)}{A(\alpha)} - r\frac{B'(\alpha)}{B(\alpha)}. \tag{14}$$

To see why the maximum is unique, write $A(\alpha) = \sum_l w_l z_l^\alpha$. The second derivative of $\log A$ is a weighted variance of $\log z_l$ and is nonnegative. Also $\log B$ is linear in α. Hence $\ell_p''(\alpha) < 0$. Distinct records make the score eventually negative; it is positive near zero. The estimate is a finite interior maximum under the stated conditions. The reliability estimate is $\widehat{R} = R(\widehat{a}, \widehat{b})$.

2.2 MLE under order restriction

Under $a \le b$, keep the rates in (12) if they satisfy the order. Otherwise the maximum is on $a = b$. Differentiating on this boundary gives the pooled rate $q/(A + B)$. Thus, at every trial shape,

$$(\widetilde{a}, \widetilde{b}) = \begin{cases} (d/A, r/B), & d/A \le r/B, \\ (q/(A+B), q/(A+B)), & d/A > r/B. \end{cases} \tag{15}$$

We maximize the resulting profile again over α and calculate $\widetilde{R} = R(\widetilde{a}, \widetilde{b})$. Sorting the two unrestricted rates would not give the correct estimate because strength and stress play different roles. Pooling only at the unrestricted shape would also omit the required reoptimization. On the pooled branch the profile curvature is $-q/\alpha^2 - q\{\log(A + B)\}'' < 0$, and the score agrees across the transition between branches.

2.3 Approximate confidence intervals

With derivatives taken with respect to α, the observed information in the order (α, a, b) is

$$I = \begin{pmatrix} q/\alpha^2 + aA'' + bB'' & A' & B' \\ A' & d/a^2 & 0 \\ B' & 0 & r/b^2 \end{pmatrix}. \tag{16}$$

The derivatives are $A' = \sum_l w_l z_l^\alpha \log z_l$, $A'' = \sum_l w_l z_l^\alpha (\log z_l)^2$, $B' = B \log y_r$, and $B'' = B(\log y_r)^2$. At the unrestricted MLE let $V = I^{-1}$. A positive approximate 95 percent interval for a parameter θ_j has endpoints

$$\exp\left[\log \widehat{\theta}_j \;\pm\; 1.96 \frac{\sqrt{V_{jj}}}{\widehat{\theta}_j}\right]. \tag{17}$$

For reliability use $\psi = \log(b/a)$ with gradient $g = (0, -1/a, 1/b)^\mathsf{T}$. Its estimated standard error is $\sqrt{g^\mathsf{T} V g}$. Exponentiate the endpoints of the normal interval for ψ to obtain ρ_L, ρ_U, then transform them to $[R(1, \rho_L), R(1, \rho_U)]$. This keeps reliability endpoints in $[0, 1]$.

A pooled ordered MLE lies on a boundary, where an ordinary normal approximation is unsuitable. We use percentile parametric bootstrap intervals for it. Each bootstrap replicate regenerates the full strength blocks, withdrawals and stopping, along with the stress records, and refits the ordered model. These intervals are finite-sample approximations. A separate simulation checks their coverage rather than assuming that it is exactly 95 percent.

3 Bayesian Estimation

3.1 Bayesian estimation under no order restriction

We use the gamma-Dirichlet construction described by Sultana et al. [1]. Put $t = a + b$ and $p = a/(a + b)$. Assign independent priors

$$t \sim \text{Gamma}(a_0, b_0), \quad p \sim \text{Beta}(c_1, c_2), \quad \alpha \sim \text{Gamma}(a_1, b_1), \tag{18}$$

where Gamma uses shape and rate. Since $|\partial(a, b)/\partial(t, p)| = t$, the induced joint rate prior is

$$\pi(a, b) = C_0 e^{-b_0(a+b)}(a + b)^{a_0-c_1-c_2} a^{c_1-1} b^{c_2-1}, \tag{19}$$
$$C_0 = \frac{b_0^{a_0}\Gamma(c_1 + c_2)}{\Gamma(a_0)\Gamma(c_1)\Gamma(c_2)}.$$

We choose $(a_0, b_0, c_1, c_2, a_1, b_1) = (2, 0.5, 2, 2, 2, 1)$ before the numerical experiments. This is a proper prior, not a noninformative prior. It gives independent rates only in the special case $a_0 = c_1 + c_2$, which is not our choice. If $c = c_1 + c_2$,

$$\text{Cov}(a, b) = \frac{a_0 c_1 c_2 (c - a_0)}{b_0^2 c^2 (c + 1)} = 0.8. \tag{20}$$

The prior means of a, b and α are all two. The same hyperparameters are used for every simulation scenario; they are not fitted to the true simulation rates.

Multiplying the likelihood and prior, and including the transformation Jacobian, gives

$$\pi(\alpha, t, p \mid D) \propto \alpha^{q+a_1-1} e^{(\alpha-1)C-b_1\alpha} p^{d+c_1-1}(1 - p)^{r+c_2-1}$$
$$\times\, t^{q+a_0-1} e^{-tH(\alpha,p)}, \tag{21}$$
$$H(\alpha, p) = b_0 + pA(\alpha) + (1 - p)B(\alpha).$$

Therefore $t \mid \alpha, p, D \sim \text{Gamma}(q + a_0, H)$. Integrating out t gives

$$\pi(\alpha, p \mid D) \propto \alpha^{q+a_1-1} e^{(\alpha-1)C-b_1\alpha} p^{d+c_1-1}(1 - p)^{r+c_2-1} H^{-(q+a_0)}. \tag{22}$$

We integrate (22) numerically over $\alpha > 0$ and $0 < p < 1$. Specifically, Gauss-Legendre quadrature is applied to $\log \alpha \in [\log(0.02), \log(30)]$ and p on its support, including the log-shape Jacobian. Small edge weights and finer-grid comparisons check the numerical bounds. Conditional expectations of t give

$$E(a \mid D) = E\left[\frac{p(q + a_0)}{H} \mid D\right], \qquad E(b \mid D) = E\left[\frac{(1 - p)(q + a_0)}{H} \mid D\right]. \tag{23}$$

Under squared error loss the reliability estimate is $\widehat{R}_B = E[R(p, 1-p) \mid D]$, not reliability evaluated at the posterior mean rates. The notebook also reports LINEX estimates. Under loss $L(\delta, R) = e^{c(\delta-R)} - c(\delta - R) - 1$, differentiation of posterior expected loss gives

$$\widehat{R}_{B,c} = -\frac{1}{c} \log E(e^{-cR} \mid D), \qquad c \neq 0. \tag{24}$$

We use $c = 1.5$ and $c = -1.5$ as supplementary calculations.

3.2 Bayesian estimation under order restriction

The restriction $a \leq b$ is equivalent to $p \leq 1/2$. We condition the joint prior on this event:

$$\pi_O(\alpha, a, b) = \frac{\pi(\alpha, a, b)\, \mathbf{1}\{a \leq b\}}{I_{1/2}(c_1, c_2)}. \tag{25}$$

For the symmetric Beta prior used here, the denominator is $1/2$. The posterior kernel is still (22), but its support is $\alpha > 0$, $0 < p \leq 1/2$. We compute posterior means on that support. We do not obtain the ordered posterior by sorting draws from the unrestricted posterior.

3.3 Credible intervals and numerical checks

An equal-tail interval uses the posterior 0.025 and 0.975 quantiles of R. A highest posterior density (HPD) set contains points with density above a threshold and total posterior probability 0.95. For a unimodal marginal it is a shortest interval of that probability.

For a numerical HPD interval, first transform the marginal density of p to the reliability scale using $f_R(R(p)) = f_p(p)/|R'(p)|$. Interpolate the density with a shape-preserving cubic function, integrate it, and minimize the length of an interval containing probability 0.95. Both ends of the support are included as candidate endpoints. We check probability mass against 0.95 and compare endpoints on a finer quadrature grid. These are numerical HPD approximations; they are not exact closed-form intervals. If the density were multimodal, a single shortest interval would not necessarily be the full HPD set.

The distinction matters at an order boundary. A continuous ordered equal-tail interval starts above 1/2, so it cannot cover a true reliability of exactly 1/2. An ordered HPD interval can start at 1/2. Under a false restriction the true reliability is below 1/2, outside the support of both interval constructions.

The simulation uses 80×110 quadrature nodes and the data examples use 240×320. Application results are checked using 400×520 nodes. As a further check, the notebook compares quadrature with four Metropolis-Hastings chains and reports traces, effective sample sizes and split R-hat. These are numerical checks on the posterior calculations, not extra estimation methods in the main comparison.

4 Numerical Illustration

4.1 Simulation study

There are 48 scenarios: three rate-order regimes, two shapes, two block sizes, two censoring protocols and two record orders. The rate pairs are $(0.8, 1.6)$, $(1, 1)$ and $(1.6, 0.8)$, called ordered, boundary and violated. Shapes are 0.8 and 1.8. Each experiment has three strength blocks of size 20 or 40, target $m = \lfloor 0.7n \rfloor$, and balanced planned withdrawals. The time limit is fixed at the true strength distribution's 65th percentile before generating the data. Both Type-I and adaptive Type-II progressive hybrid protocols are considered. The record order is $k = 1$ or $k = 3$, with six records for $n = 20$ and twelve for $n = 40$.

Each scenario has $M = 300$ independent replications, giving 14,400 datasets. All four estimators use the same data within a replication. Increasing the size also increases the number of stress records, so it does not isolate the effect of strength sample size. A fixed master seed of 250150045 makes the calculations reproducible.

If $e_l = \widehat{\theta}_l - \theta$ is the error in replication l, we calculate

$$\mathrm{AB} = \left| \frac{1}{M} \sum_{l=1}^{M} e_l \right|, \qquad \mathrm{MSE} = \frac{1}{M} \sum_{l=1}^{M} e_l^2. \tag{26}$$

Here AB is absolute bias, not mean absolute error. For intervals $[L_l, U_l]$, coverage probability (CP) and average width (AW) are

$$\widehat{\mathrm{CP}} = \frac{1}{M} \sum_{l=1}^{M} \mathbf{1}\{L_l \le \theta \le U_l\}, \qquad \mathrm{AW} = \frac{1}{M} \sum_{l=1}^{M} (U_l - L_l). \tag{27}$$

The estimated Monte Carlo standard errors are

$$\mathrm{MCSE}(\mathrm{Bias}) = \frac{\mathrm{SD}(e_l)}{\sqrt{M}}, \tag{28}$$

$$\mathrm{MCSE}(\widehat{\mathrm{CP}}) = \sqrt{\frac{\widehat{\mathrm{CP}}(1-\widehat{\mathrm{CP}})}{M}}. \tag{29}$$

They are saved with the scenario-level results. The notebook and CSV tables also contain parameter errors for α, a, b, as well as RMSE. The manuscript tables focus on the assigned system reliability.

4.1.1 Generating one dataset

Algorithm 1 describes the data-generation part. All initial strength lifetimes are generated first, but only failures and censoring information that would be observed under the chosen experiment are passed to the estimators. A withdrawn unit's actual lifetime is not used later in the likelihood. This prevents the estimation step from receiving information that the sampling scheme was meant to hide.

For stress records we work on the cumulative-hazard scale. If E_j are independent exponential variables with mean one, the generated record values are

$$y_j = \left(\frac{\sum_{l=1}^{j} E_l}{kb} \right)^{1/\alpha}, \qquad j = 1, \ldots, r. \tag{30}$$

This generates the marginal record values directly. It does not simulate or use inter-record waiting times. In the published-data examples we instead scan the chosen observation order and retain its actual upper k-records.

4.1.2 Finding estimates and comparing them

Algorithm 2 describes what is found from each generated dataset. The truth is fixed within a scenario. It is not drawn from the Bayesian prior. The same observations are used for unrestricted MLE, ordered MLE, unrestricted Bayes and ordered Bayes.

The ordered MLE does not receive an ordinary normal interval at a pooled boundary. Its bootstrap comparison is separate and has fewer outer replications. Missing interval entries in the main ordered MLE table therefore do not mean zero coverage. They mean that the regular asymptotic interval was not used for that estimator.

4.1.3 Posterior and bootstrap calculations

Algorithm 3 gives the numerical Bayesian steps. Integrating out the total rate t first leaves only two dimensions. This is why the calculation can be repeated for all simulation settings without fitting an additional high-dimensional model.

For an ordered MLE bootstrap interval, we first fit the ordered model. We then generate a new complete experiment under that fit, apply the same censoring and record rules, and refit it. After repeating this step, the empirical 0.025 and 0.975 quantiles of bootstrap reliability estimates form the percentile interval. The real-data examples use 500 resamples; the outer coverage check uses 199 inner resamples for each of its 100 outer datasets. Reusing a complete-data iid bootstrap without the censoring and record steps would represent a different observation scheme.

Algorithm 1 Generate a censored strength sample and stress records

Input: True (α, a, b); block sizes, targets, thresholds, withdrawals; protocol; record order k and count r

Output: Observed failures, actual withdrawals, terminal counts and r stress records

for each strength block i **do**
  Draw n_i independent values U uniformly on $(0, 1)$.
  Set each initial lifetime to $[-\log(1-U)/a]^{1/\alpha}$.
  Mark all starting specimens as active; set the failure count to zero.
  while the target number of failures has not been reached **do**
    Find the smallest lifetime among active specimens.
    if the protocol is Type I and this lifetime exceeds T_i **then**
      Censor all active specimens at T_i and stop this block.
    else
      Record the failure and remove that specimen from the active set.
      if this failure reaches the target **then**
        Withdraw all remaining active specimens and stop this block.
      else
        Use the planned withdrawal count; for adaptive Type II, use zero after T_i.
        Choose that many active survivors uniformly and withdraw them.
      end if
    end if
  end while
  Save the actual withdrawal counts and any terminal survivors.
end for
Draw r independent exponential variables $E_1, \ldots, E_r$ of mean one.
Form y_j from equation (30).
Check specimen conservation and strictly increasing stress records.
Return the retained observations, without hidden withdrawn lifetimes.

Algorithm 2 Monte Carlo comparison of the four estimators

Input: The 48 predeclared settings, $M = 300$, fixed prior and master seed
Output: Point errors, interval coverage, interval widths and saved raw results

for each scenario **do**
  Calculate true reliability from equation (6).
  Start the scenario's reproducible random-number stream.
  for $l = 1, \ldots, M$ **do**
    Generate one dataset by Algorithm 1.
    Construct $A(\alpha)$, $B(\alpha)$, C, d and r using retained data.
    Maximize the unrestricted profile and calculate its rates and reliability.
    Maximize the ordered profile using pooling whenever the rates violate the order.
    Integrate the unrestricted posterior and obtain means, equal-tail and HPD intervals.
    Repeat posterior integration on the ordered support.
    Form the unrestricted log-ratio asymptotic reliability interval.
    Save point estimates, interval endpoints and indicators of coverage.
    Save the retained failure count, pooling status and numerical edge weights.
  end for
  Calculate absolute bias, MSE and RMSE for α, a, b and R.
  Calculate coverage and average width for each reported reliability interval.
  Calculate the Monte Carlo standard errors of the performance estimates.
end for
Export scenario settings, raw replications and summary tables.
Assess ordered MLE bootstrap coverage separately in six selected settings.

Algorithm 3 Compute Bayesian estimates and reliability intervals

Input: Retained observations, joint prior and choice of ordered or unrestricted support
Output: Posterior means and 95 percent equal-tail and numerical HPD intervals

Choose quadrature nodes for $\log \alpha$ and p on the stated support.
At every node calculate A, B, H and the collapsed log posterior.
Include quadrature weights and the Jacobian for $\log \alpha$.
Normalize weights using a log-sum-exp calculation.
Average α, conditional rate means and $R(p, 1-p)$ with these weights.
Obtain the reliability marginal and its equal-tail quantiles.
Transform the marginal density from p to R using $|dR/dp|$.
Interpolate and integrate this density on its reliability support.
Find the shortest interval of probability 0.95, also checking support endpoints.
Check interval probability mass and compare selected fits on a finer grid.

Table 2: Absolute bias and MSE of reliability, averaged over 16 scenarios in each rate regime.

Rate regime	Estimator	Absolute bias	MSE
Ordered	MLE	0.0078	0.0130
Ordered	Ordered MLE	0.0069	0.0117
Ordered	Bayes	0.0283	0.0099
Ordered	Ordered Bayes	0.0050	0.0051
Boundary	MLE	0.0089	0.0171
Boundary	Ordered MLE	0.0550	0.0097
Boundary	Bayes	0.0107	0.0115
Boundary	Ordered Bayes	0.1076	0.0142
Violated	MLE	0.0070	0.0126
Violated	Ordered MLE	0.1969	0.0393
Violated	Bayes	0.0345	0.0102
Violated	Ordered Bayes	0.2501	0.0632

4.1.4 Results for the model parameters

Reliability depends on the rate ratio, but the common shape and individual rates are also estimated from the retained observations. The next three tables therefore report their point-estimation errors. These are different targets with different scales. An MSE for a should not be compared numerically with an MSE for R as if both measured the same quantity.

Table 3: Point-estimation errors for α, averaged over the 16 scenarios in each rate regime.

Regime	Estimator	Absolute bias	MSE	RMSE
Ordered	MLE	0.0400	0.0300	0.1556
Ordered	Ordered MLE	0.0384	0.0289	0.1531
Ordered	Bayes	0.0421	0.0274	0.1503
Ordered	Ordered Bayes	0.0328	0.0250	0.1434
Boundary	MLE	0.0429	0.0296	0.1547
Boundary	Ordered MLE	0.0167	0.0222	0.1340
Boundary	Bayes	0.0354	0.0252	0.1442
Boundary	Ordered Bayes	0.0272	0.0205	0.1301
Violated	MLE	0.0478	0.0299	0.1562
Violated	Ordered MLE	0.0936	0.0268	0.1516
Violated	Bayes	0.0260	0.0226	0.1375
Violated	Ordered Bayes	0.1235	0.0339	0.1696

All table entries first summarize the 300 replications within a scenario and then average over the 16 scenarios in a regime. Thus the root mean squared error column is an average of scenario RMSE values, rather than the square root of the displayed average MSE. Detailed results retain each scenario separately. Averaging is useful for a short comparison, but it can hide variation between shapes, sizes and sampling designs.

The order assumption acts directly on the two rates. When it is false, pooling or truncation changes their estimates and can also change the fitted common shape. A prior may reduce a variance while shifting an estimate away from the truth. This is why the comparison uses MSE as well as bias and avoids calling one method best for every target.

For the ordered rate regime, mean scenario-specific reliability MSE is 0.0130 for unrestricted MLE, 0.0099 for unrestricted Bayes and 0.0051 for ordered Bayes.

For the boundary rate regime, mean scenario-specific reliability MSE is 0.0171 for unrestricted

Table 4: Point-estimation errors for a, averaged over the 16 scenarios in each rate regime.

Regime	Estimator	Absolute bias	MSE	RMSE
Ordered	MLE	0.0137	0.0160	0.1236
Ordered	Ordered MLE	0.0128	0.0157	0.1226
Ordered	Bayes	0.0261	0.0163	0.1246
Ordered	Ordered Bayes	0.0176	0.0147	0.1187
Boundary	MLE	0.0219	0.0257	0.1570
Boundary	Ordered MLE	0.0065	0.0205	0.1409
Boundary	Bayes	0.0296	0.0248	0.1543
Boundary	Ordered Bayes	0.0242	0.0187	0.1347
Violated	MLE	0.0694	0.0912	0.2920
Violated	Ordered MLE	0.1790	0.0743	0.2676
Violated	Bayes	0.0463	0.0746	0.2659
Violated	Ordered Bayes	0.2298	0.0907	0.2949

Table 5: Point-estimation errors for b, averaged over the 16 scenarios in each rate regime.

Regime	Estimator	Absolute bias	MSE	RMSE
Ordered	MLE	0.1884	0.6600	0.7869
Ordered	Ordered MLE	0.1938	0.6513	0.7819
Ordered	Bayes	0.1096	0.3551	0.5886
Ordered	Ordered Bayes	0.1829	0.3262	0.5634
Boundary	MLE	0.1114	0.2885	0.5151
Boundary	Ordered MLE	0.2317	0.2537	0.4788
Boundary	Bayes	0.1580	0.2125	0.4493
Boundary	Ordered Bayes	0.4331	0.3051	0.5398
Violated	MLE	0.0712	0.1614	0.3920
Violated	Ordered MLE	0.6448	0.4781	0.6893
Violated	Bayes	0.1723	0.1585	0.3905
Violated	Ordered Bayes	0.8636	0.8221	0.9031

MLE, 0.0115 for unrestricted Bayes and 0.0142 for ordered Bayes.

For the violated rate regime, mean scenario-specific reliability MSE is 0.0126 for unrestricted MLE, 0.0102 for unrestricted Bayes and 0.0632 for ordered Bayes.

4.1.5 Information retained under each protocol

The starting number of specimens is not the observed failure count. Type-I stopping can occur before the target. Adaptive Type II continues until its target is reached. Both experiments include censoring information in their exposure terms, but they do not retain the same failures or have the same duration. The histograms in the simulation figures show this difference directly. Under

Table 6: Retained strength information across the simulation. Means pool all rate and shape settings at the indicated size and protocol. The target is the sum across three blocks.

Protocol	n	Target	Mean failures	SD	Time-stopped blocks
Adaptive II	20	42	42.00	0.00	0.00
Adaptive II	40	84	84.00	0.00	0.00
Type I	20	42	29.02	2.78	2.99
Type I	40	84	57.80	3.98	3.00

adaptive Type II the totals are fixed at 42 or 84 failures because there are three blocks. Under Type I the failure count varies. More observed failures do not by themselves establish that one design is preferable, since test duration and withdrawals also change. Tables 2 and 7 average

Table 7: Coverage probability and average width of 95 percent reliability intervals, averaged over 16 scenarios in each regime. HPD denotes the numerical approximation described in Section 3.3.

Rate regime	Interval	CP	AW
Ordered	MLE ACI	0.941	0.401
Ordered	Bayes equal-tail	0.967	0.402
Ordered	Bayes HPD	0.955	0.392
Ordered	Ordered Bayes equal-tail	0.984	0.323
Ordered	Ordered Bayes HPD	0.956	0.313
Boundary	MLE ACI	0.938	0.453
Boundary	Bayes equal-tail	0.963	0.430
Boundary	Bayes HPD	0.941	0.427
Boundary	Ordered Bayes equal-tail	0.000	0.253
Boundary	Ordered Bayes HPD	0.914	0.233
Violated	MLE ACI	0.940	0.405
Violated	Bayes equal-tail	0.959	0.386
Violated	Bayes HPD	0.946	0.380
Violated	Ordered Bayes equal-tail	0.000	0.174
Violated	Ordered Bayes HPD	0.000	0.151

the 16 scenario-specific summaries within each rate regime. In particular, the reported AB is the average of the scenario-specific absolute biases. It is not the absolute value of an average signed bias.

The results show why an order restriction must be justified. In the ordered regime, the ordered Bayesian estimator has a smaller reliability MSE than the unrestricted methods in this experiment. In the violated regime, the restricted methods cannot estimate reliability below one half and have much larger bias. There is no general advantage to restricting the model when the assumed order is false.

At the equality boundary, ordered equal-tail intervals have coverage 0.000, whereas ordered numerical HPD intervals have coverage 0.914. Their mean widths are 0.253 and 0.233, respectively. The HPD calculation allows the lower endpoint to equal one half. This explains why its boundary coverage differs from the equal-tail result; it does not establish nominal coverage in every scenario.

4.1.6 Reading the simulation graphs

Figures 4–6 show how the reliability errors change with the size setting. Each plotted point averages eight scenarios at that size. The upper panel is MSE and the lower panel is signed bias. A point closer to zero in the bias panel does not necessarily have the lowest MSE, because MSE also reflects variation across datasets.

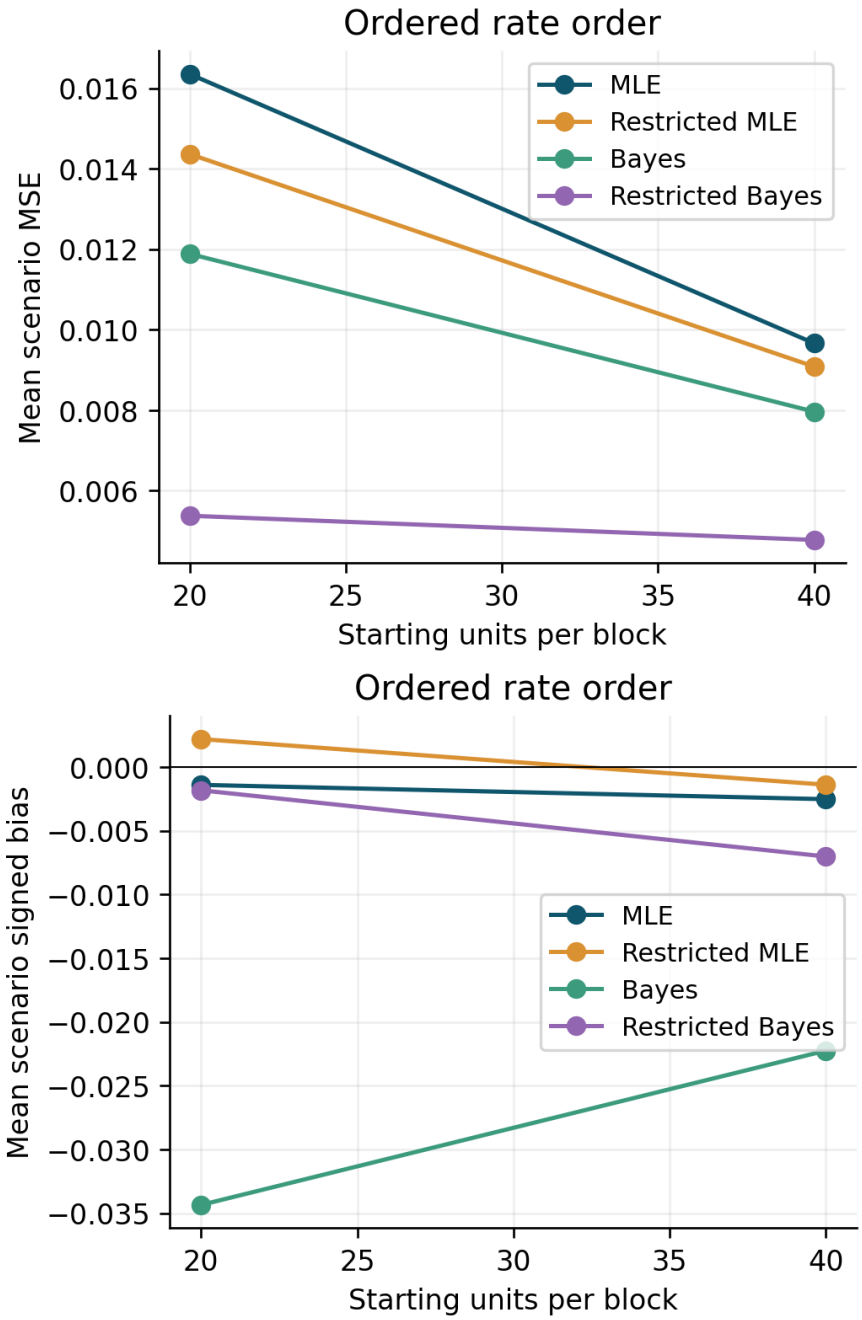


Figure 4: Reliability MSE and signed bias in the ordered rate regime. Both the strength size and stress record count increase between the two horizontal-axis values.

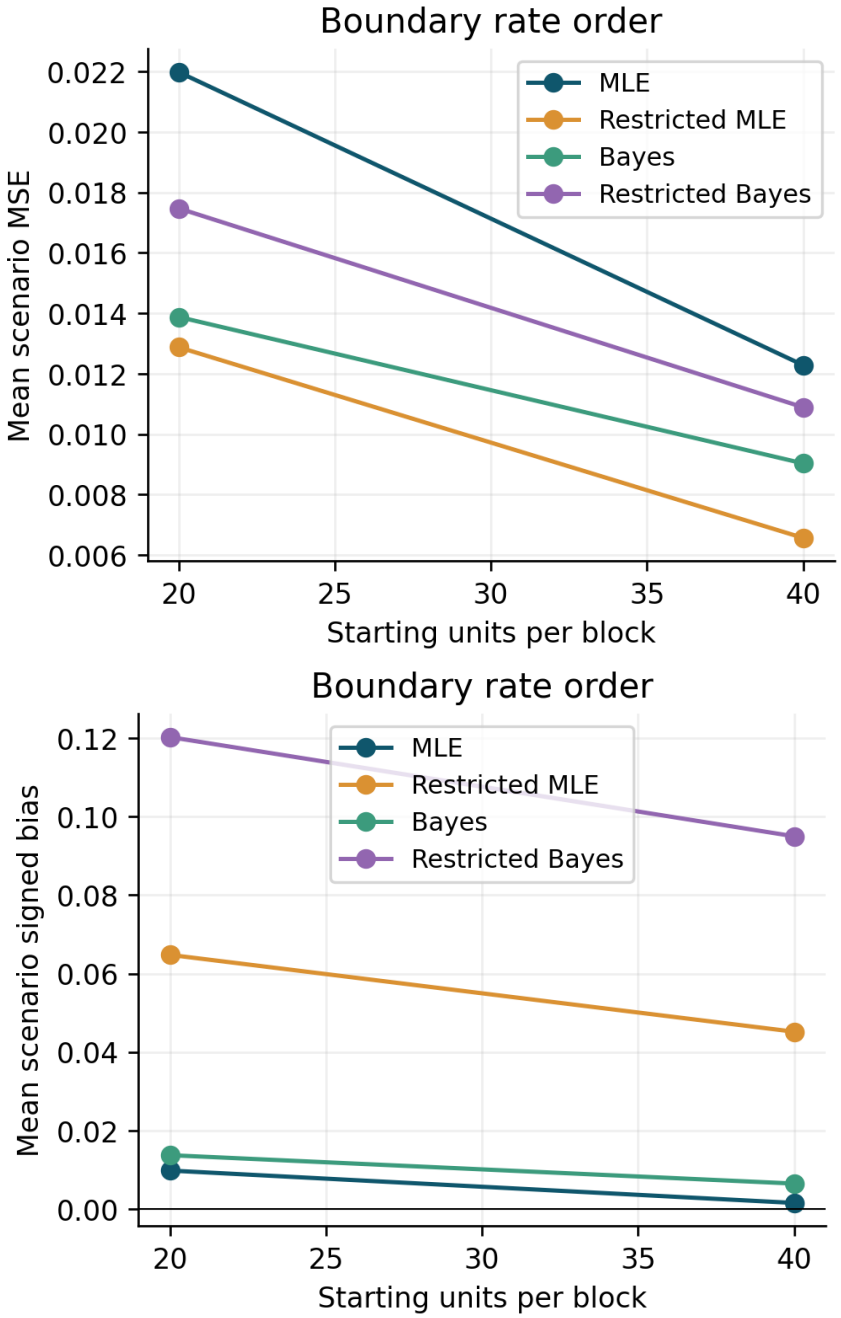


Figure 5: Reliability MSE and signed bias in the boundary rate regime. Both the strength size and stress record count increase between the two horizontal-axis values.

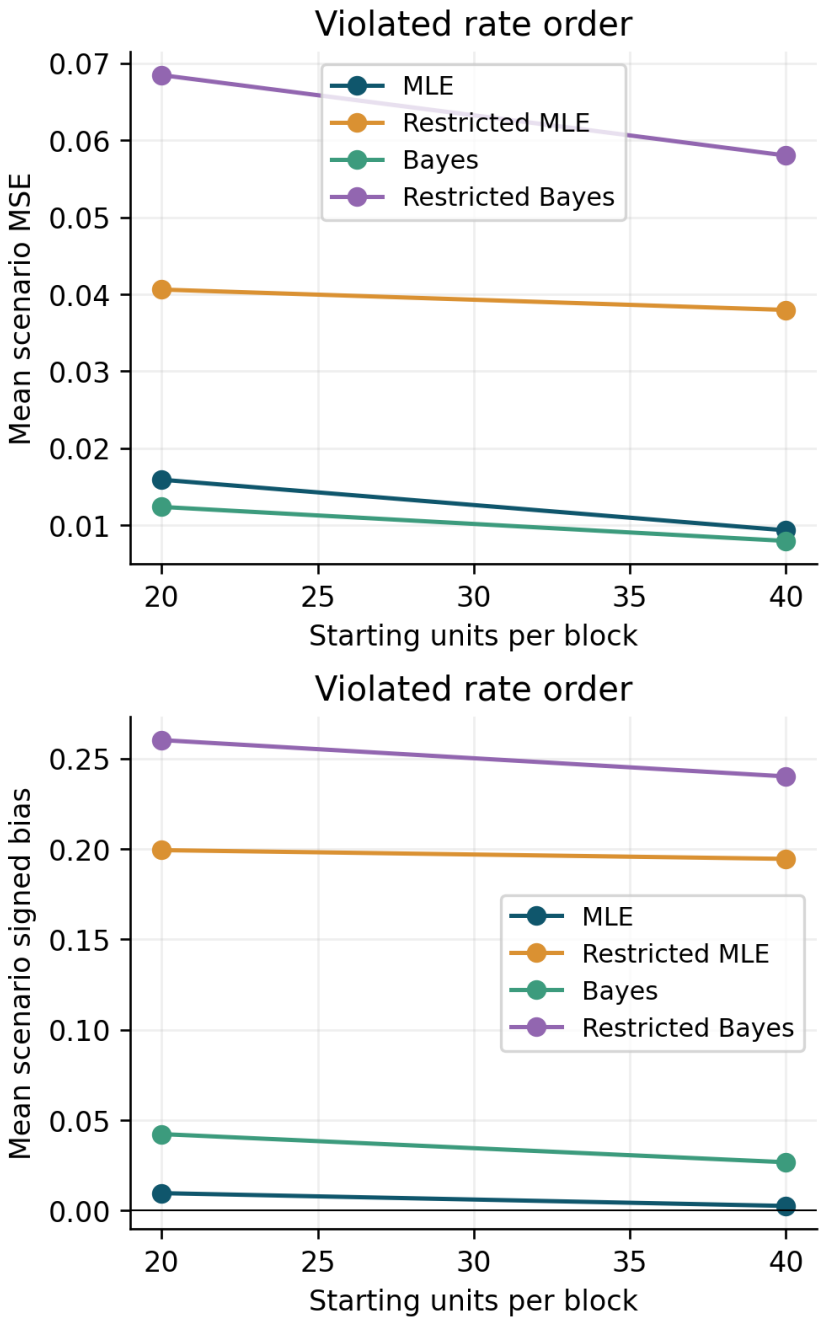


Figure 6: Reliability MSE and signed bias in the violated rate regime. Both the strength size and stress record count increase between the two horizontal-axis values.

In the ordered cases the order restriction can reduce uncertainty by excluding rate pairs that are assumed impossible. At the boundary, it tends to shift reliability upward. In the violated cases, a restricted reliability estimate cannot approach the true value below one half, even if more observations are collected. The bias in that case is a consequence of the assumption rather than a numerical fitting failure.

The coverage plots display individual scenario results instead of only regime averages. Error bars are 1.96 times the estimated Monte Carlo standard error, and the horizontal reference is 0.95. The bars describe uncertainty in a simulated coverage estimate; they are not confidence intervals for the reliability of an individual specimen.

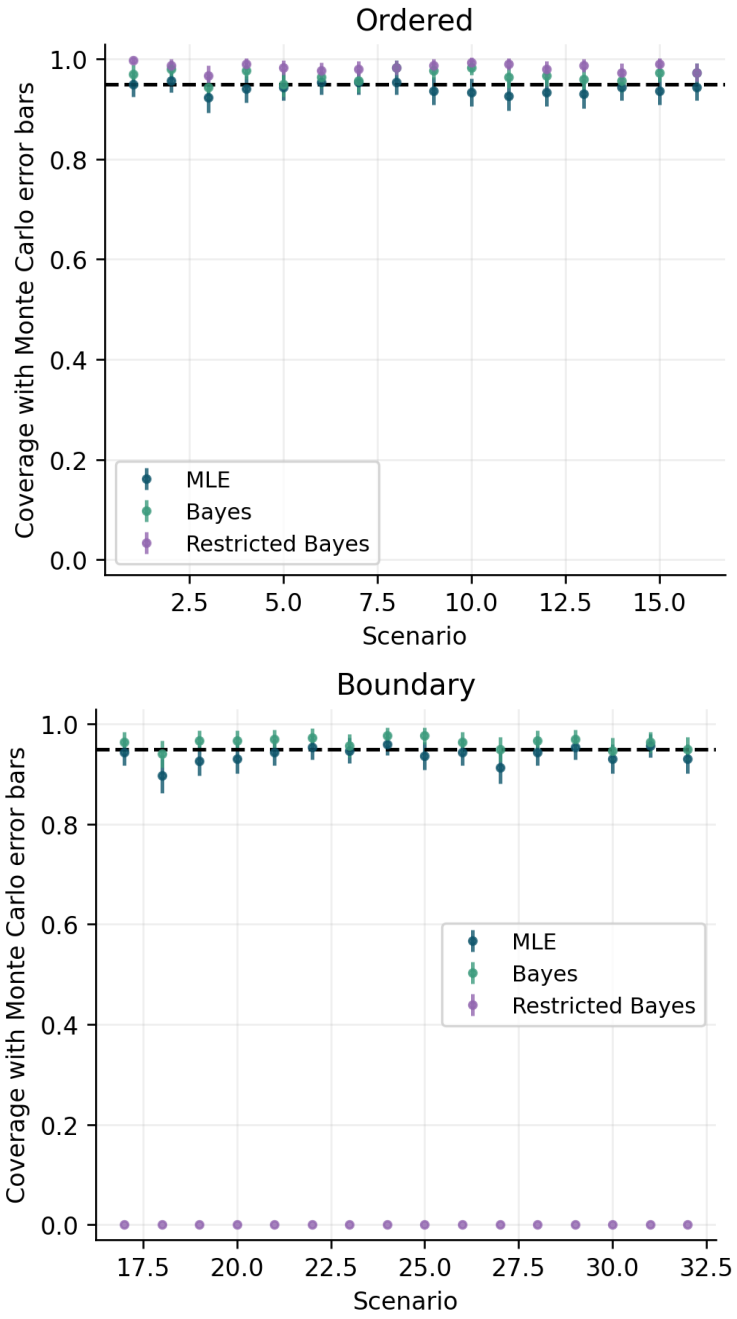


Figure 7: Scenario-level coverage of the unrestricted ACI and the two Bayesian equal-tail intervals in the ordered and boundary regimes.

The HPD comparison makes the boundary distinction visible. An ordered equal-tail interval starts

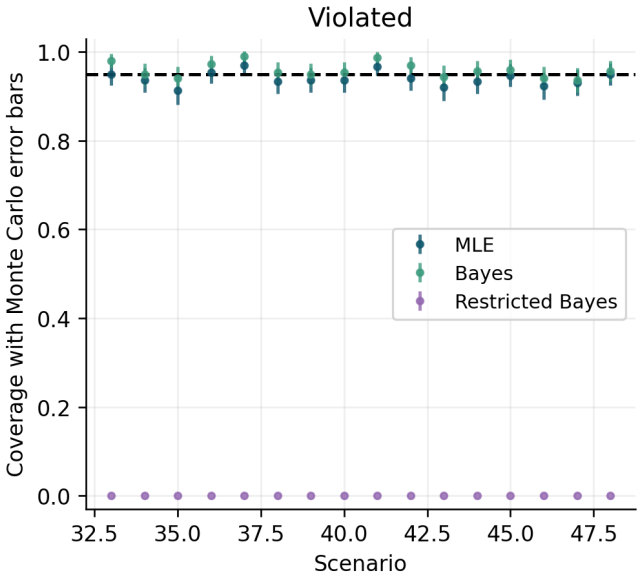


Figure 8: Scenario-level coverage in the violated regime. Ordered equal-tail intervals cannot cover a truth below their reliability support.

strictly above one half, whereas a shortest interval can use one half as an endpoint. Shorter numerical HPD intervals therefore need to be assessed by coverage as well as width. The reported averages are not a claim that every scenario attains nominal coverage.

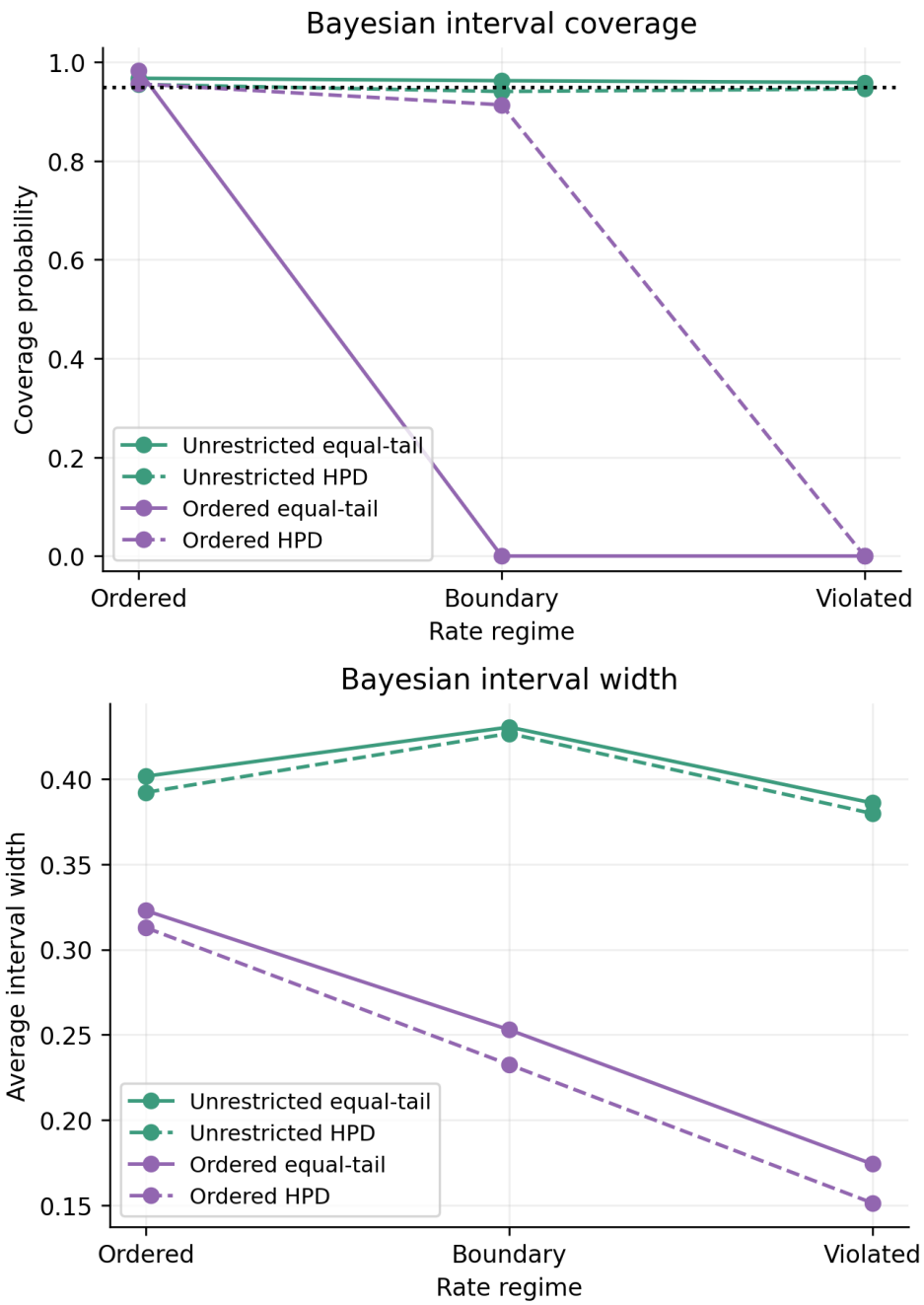


Figure 9: Coverage and average width for unrestricted and ordered Bayesian equal-tail and numerical HPD intervals. Each value averages 16 scenario summaries.

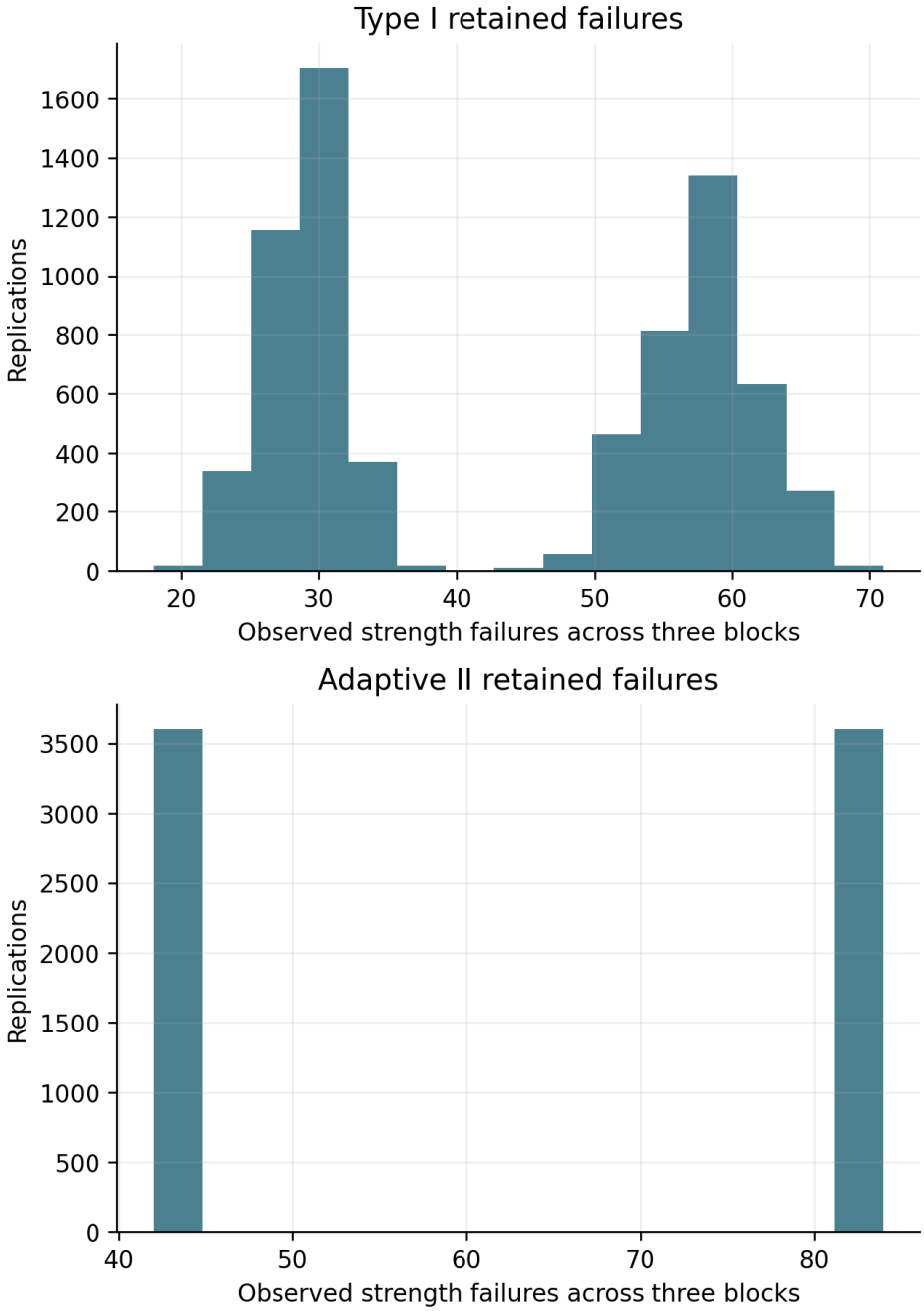


Figure 10: Observed strength failure totals under Type-I and adaptive Type-II progressive hybrid sampling, pooled across the corresponding simulation settings.

Ordered MLE bootstrap coverage is assessed separately in six selected scenarios, with 100 outer datasets and 199 bootstrap resamples each. This smaller comparison has larger Monte Carlo uncertainty. It does not assign a regular normal interval to a boundary estimate.

Table 8: Ordered MLE bootstrap intervals in six selected scenarios: 100 outer datasets and 199 resamples per dataset. MCSE refers to coverage.

Rate regime	Protocol	CP	MCSE	AW
Ordered	Type I	0.910	0.029	0.371
Ordered	Adaptive II	0.860	0.035	0.364
Boundary	Type I	0.940	0.024	0.340
Boundary	Adaptive II	0.910	0.029	0.343
Violated	Type I	0.000	0.000	0.321
Violated	Adaptive II	0.000	0.000	0.306

4.2 Real data analysis

The examples use published breaking-strength measurements. Both strength and comparison stress are measured in the same units within an example. The comparison group is a laboratory reference population, not a set of measured operating loads. The calculated reliability should therefore not be described as a certified in-service probability.

Table 9: Descriptive statistics for all supplied observations used in the applications, before retrospective censoring. Jute values use the source scaling; carbon values in this table are in GPa.

Sample	n	Mean	SD	Median	Minimum	Maximum
Jute 15 mm	30	0.3639	0.2430	0.3474	0.0427	0.8139
Jute 20 mm	30	0.3407	0.2473	0.2646	0.0367	0.7651
Carbon (GPa)	100	2.6214	1.0139	2.7000	0.3900	5.5600

4.2.1 Jute-fibre measurements

The 15 mm and 20 mm jute-fibre groups have 30 observations each. They are transcribed, in printed order, from Table 10 of Sultana et al. [1], where they are attributed to Xia et al. [4]. The reproduced measurements have already been divided by 1000. We use 15 mm as strength and 20 mm as comparison stress. The 10 mm group is excluded because the reproduced table contains the unverified value 70.32383. We do not guess a replacement for it. The original Xia experimental file was not available; the supplied secondary table is the transcription source.

We first fit the complete groups and assess Weibull fit using bootstrap-calibrated Kolmogorov-Smirnov tests with refitting in each of 1000 resamples. We also test the common-shape assumption by a bootstrap likelihood-ratio test. Fitted-parameter K-S tests need this calibration; ordinary distribution-free K-S p-values would not apply. For the jute groups, the common-shape bootstrap p-value is 0.620. The calibrated K-S p-values are 0.026 and 0.078. Thus the first group rejects Weibull adequacy at 5 percent, even though the common-shape test does not reject. The jute estimates must be read with this fit limitation. The carbon common-shape bootstrap p-value is 0.678. Table 10 reports its separate fit checks.

Next, a seeded allocation divides the strength group into three blocks of ten. Each block has target seven failures, balanced withdrawals and $T = 0.5$. Type-I progressive hybrid censoring leaves 13 observed failures altogether. From the comparison group we retain the first six upper 2-records,

Table 10: Complete-data Weibull fits and bootstrap-calibrated K-S tests. Carbon rates use the fixed 3 GPa measurement unit.

Sample	n	$\widehat{\alpha}$	Rate	K-S	p-value
Jute strength	30	1.499	3.908	0.169	0.026
Jute comparison	30	1.361	3.839	0.149	0.078
Carbon strength	50	2.719	1.180	0.078	0.585
Carbon comparison	50	2.897	0.942	0.113	0.101

stopping at position nine. The printed order is an imposed acquisition order; it is not known to be the historical order of the measurements. The notebook examines alternative random orders as a sensitivity check. Allocation, withdrawals, terminal survivors and record positions are saved in the data folder.

Table 11: All observed jute failures, actual withdrawals and terminal survivor counts. Terminal survivors are counted once per block, not once per displayed row; all blocks stop at time 0.5.

Block	Failure no.	Failure	Withdrawn	Terminal survivors
1	1	0.07638	1	4
1	2	0.20275	1	4
1	3	0.33922	1	4
2	1	0.07224	1	2
2	2	0.10673	1	2
2	3	0.20076	1	2
2	4	0.22565	0	2
2	5	0.46847	0	2
3	1	0.04266	1	2
3	2	0.07009	1	2
3	3	0.08040	1	2
3	4	0.13509	0	2
3	5	0.15667	0	2

Table 12: The six upper 2-records retained from the jute comparison group, with positions in printed acquisition order.

Record	k	Value	Acquisition position
1	2	0.07146	2
2	2	0.28464	3
3	2	0.41902	4
4	2	0.45660	5
5	2	0.58557	8
6	2	0.66266	9

4.2.2 Jute fit and retained-data graphs

The density plots compare histograms of the complete measurements with fitted Weibull densities. Histogram shape depends on binning, so a visual match alone is not a test. The CDF panels compare all ordered observations with fitted cumulative probabilities. The Q-Q panels show where sample quantiles differ from those predicted by the fitted model.

These complete-data checks use every measurement in the two groups. They are separate from the later retained-data likelihood. The failure plot below shows only observed failures, with ring

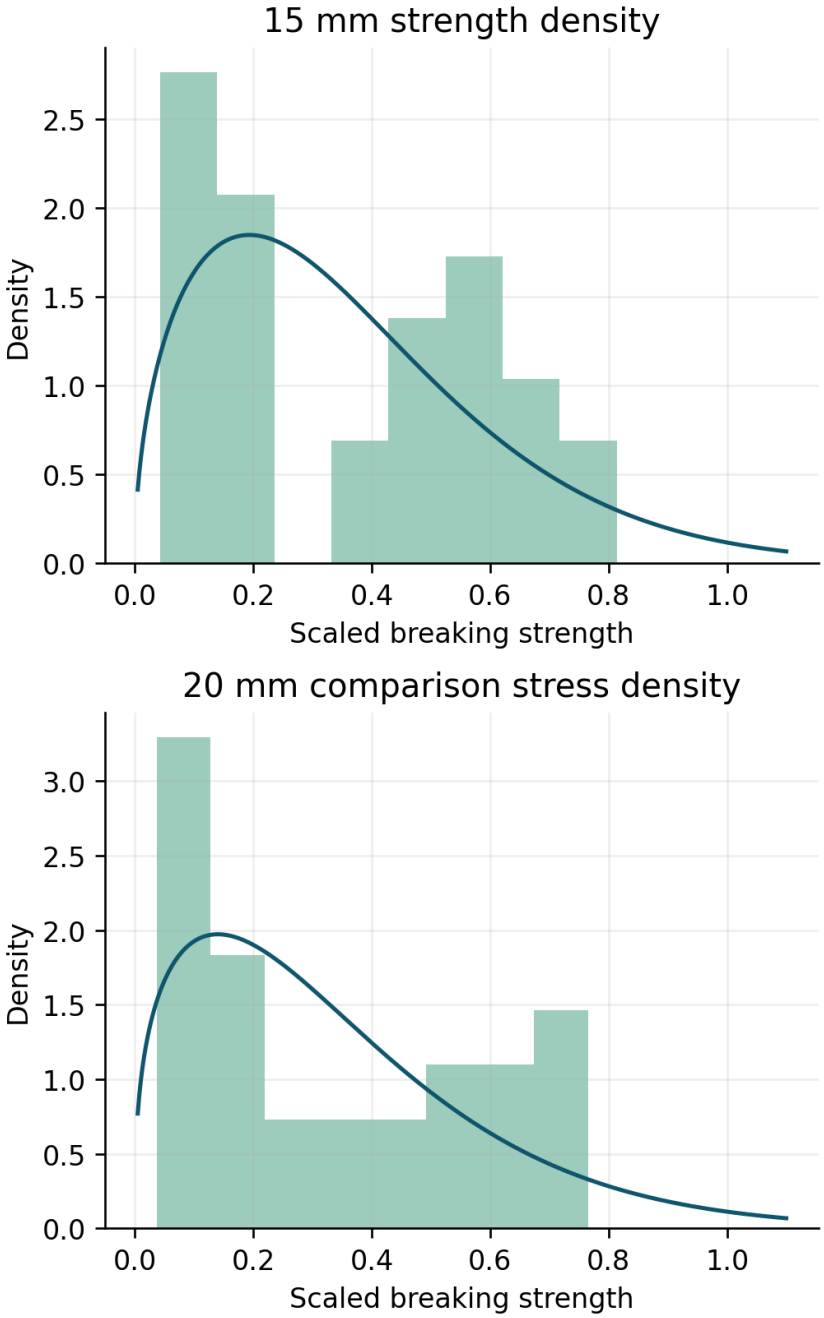


Figure 11: Jute complete-data histograms and fitted densities: 15 mm strength group above and 20 mm comparison group below.

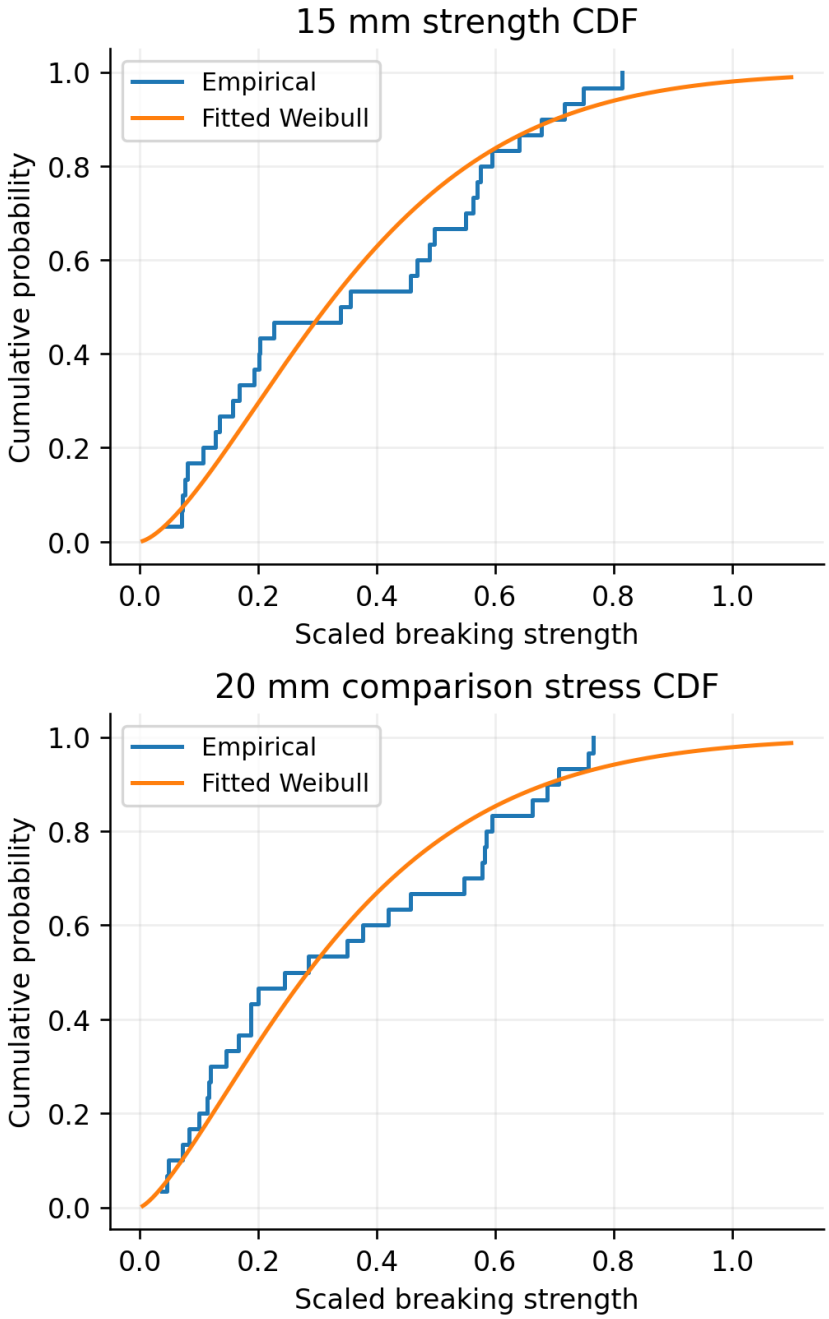


Figure 12: Jute empirical and fitted cumulative distribution functions for the same two groups.

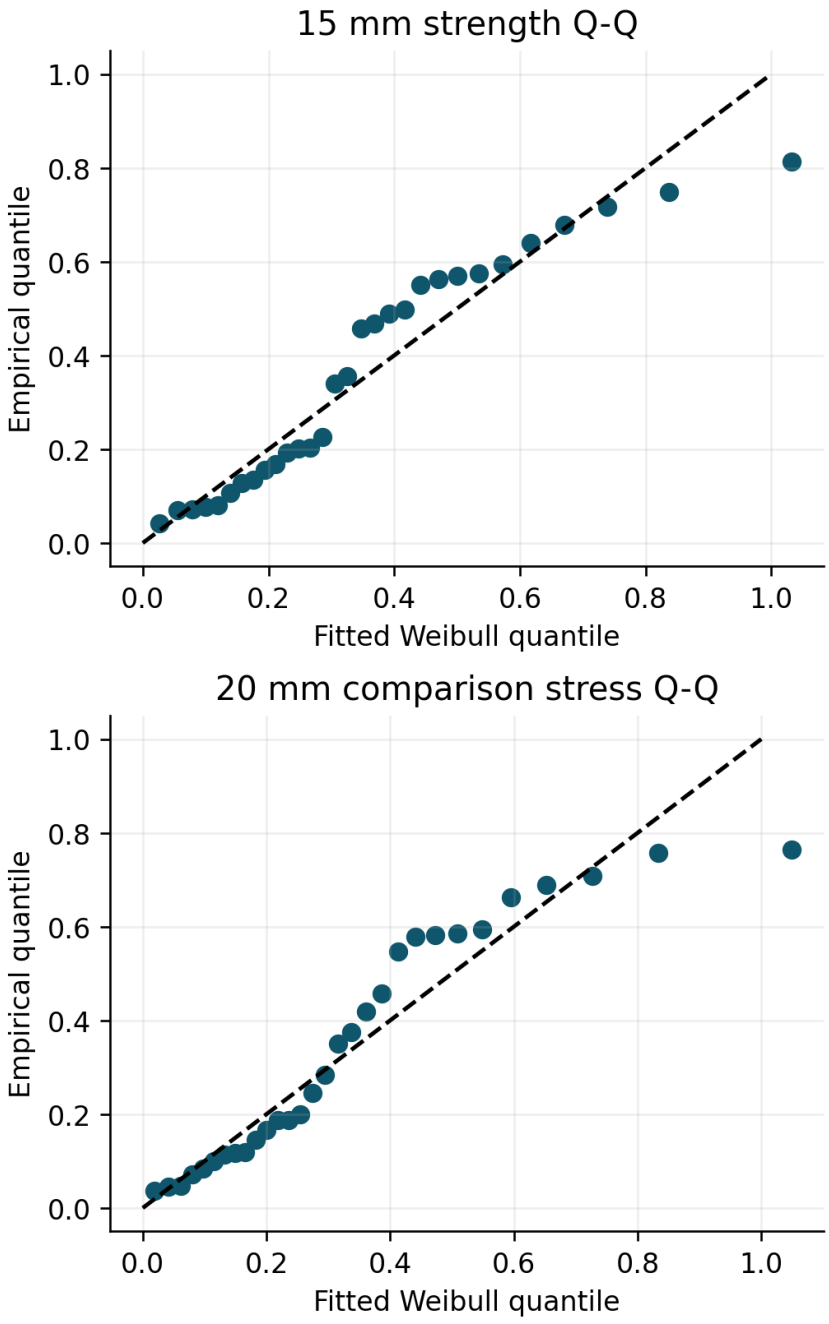


Figure 13: Jute Weibull Q-Q plots. Their deviations should be read alongside the calibrated tests, especially the rejection in the 15 mm group.

size indicating the actual number withdrawn at a failure. The terminal survivors still enter the likelihood through their survival at the time limit. The record plot shows both the complete comparison sequence and the subset actually used as upper records.

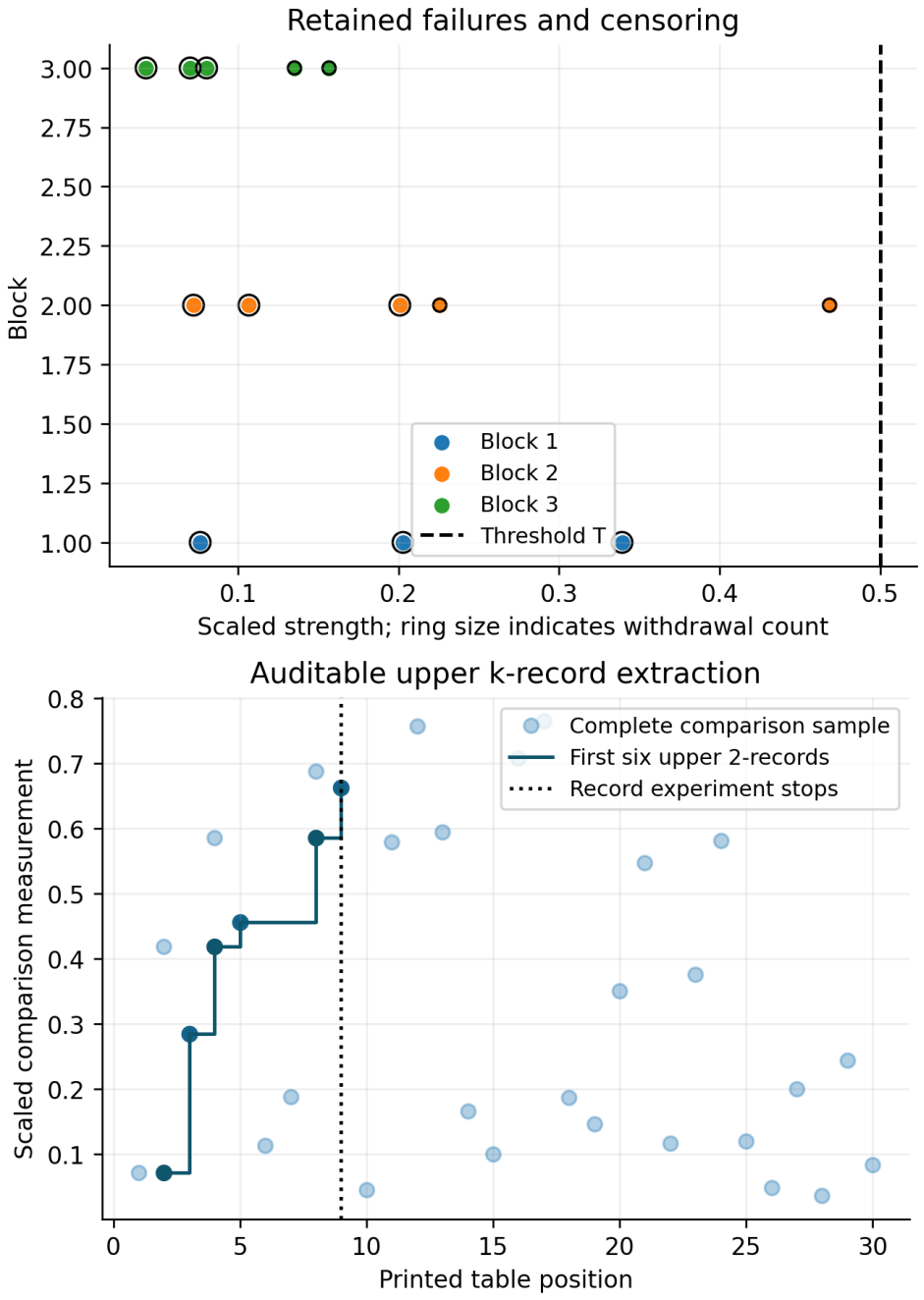


Figure 14: Retrospective jute sampling: strength failures with actual withdrawals, and extraction of the first six upper 2-records from the printed comparison ordering.

4.2.3 Additional carbon-fibre measurements

Because of the jute fit limitation, we also use the 100 carbon-fibre measurements reproduced by Newer and Alanazi [3], Section 5.1.1, attributed there to Nichols and Padgett [5]. They are in GPa. A seeded random permutation splits them into disjoint groups of fifty. We divide both groups by a fixed 3 GPa unit before inference, and keep the same prior hyperparameters on these scaled rates.

The original Nichols and Padgett experimental file was not supplied; the reproduced article is the source for our transcription.

Strength block sizes are 17, 17 and 16, with target $\lfloor 0.7n \rfloor$, $T = 1$ in scaled units and balanced withdrawals. We use the first five upper 4-records from the comparison half. The retained records are distinct, so no jitter is added to rounded source values. Both complete groups are compatible with the fitted Weibull distributions at the 5 percent level, and the common-shape test does not reject. This supports using the carbon example as an additional illustration, without proving the assumptions true.

Table 13: Retrospective carbon strength blocks under Type-I progressive hybrid censoring. The time limit is one in units of 3 GPa. Each row conserves all starting specimens.

Block	n_i	m_i	Failures	Withdrawn	Terminal
1	17	11	8	6	3
2	17	11	8	6	3
3	16	11	9	5	2

Table 14: The five upper 4-records from the carbon comparison half. Source position refers to the original table; acquisition position refers to the seeded comparison ordering.

Record	Scaled value	Source position	Acquisition position
1	0.32667	2	4
2	0.91333	58	5
3	1.06333	61	8
4	1.21667	80	14
5	1.22667	73	24

4.2.4 Carbon fit graphs

The carbon figures repeat the same density, CDF and Q-Q checks for disjoint groups obtained by the seeded split. Both groups measure the same physical quantity, and both are scaled by the fixed 3 GPa unit before inference. This keeps the comparison meaningful in units, although one group is still a reference stress population rather than measured operating loads.

The closer visual agreement and the calibrated tests support this additional illustration. They do not establish that the fitted distribution is exact or that future experiments must behave the same way. The estimates below are conditional on common shape, independent observations before selection and noninformative withdrawals.

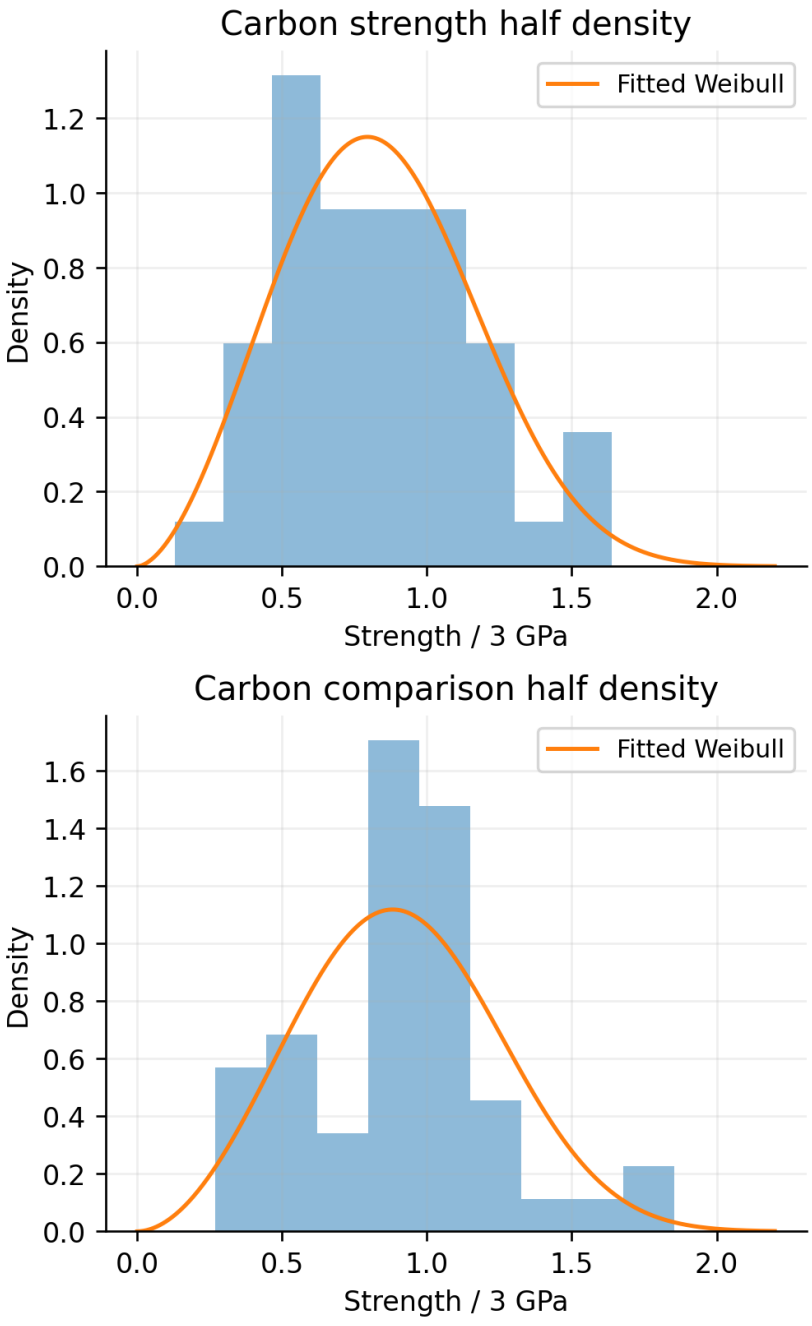


Figure 15: Carbon histograms and fitted Weibull densities for the strength and comparison halves, in units of 3 GPa.

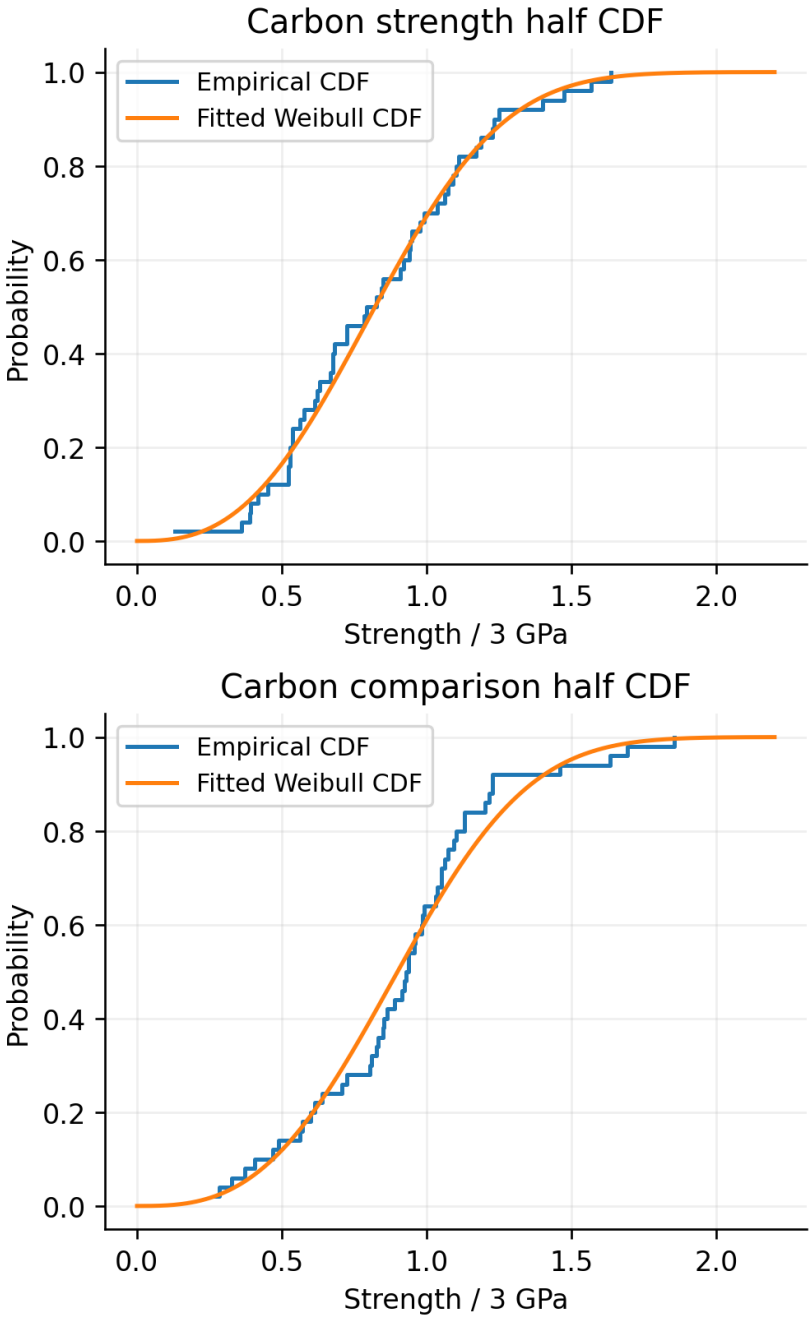


Figure 16: Carbon empirical and fitted cumulative distributions for the two allocated halves.

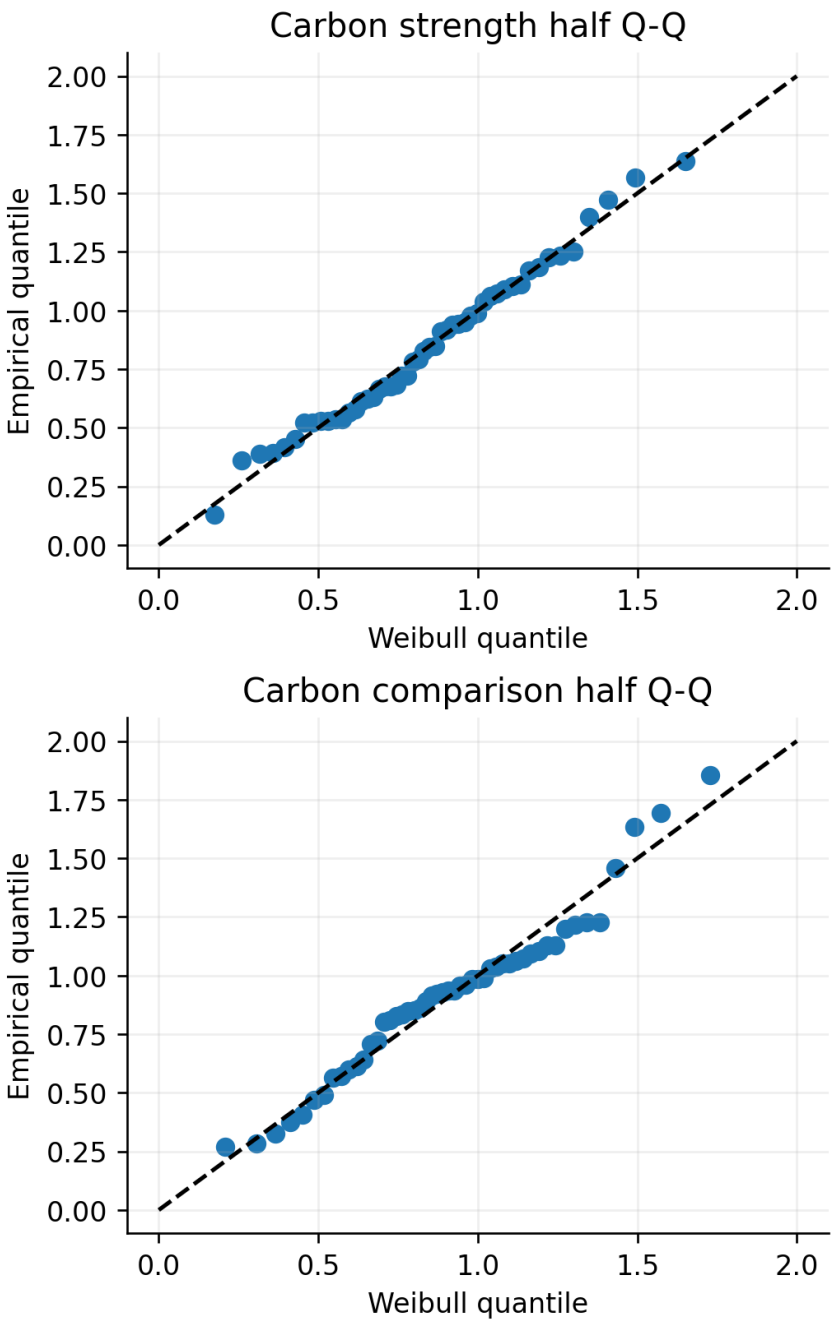


Figure 17: Carbon Weibull Q-Q plots for the two allocated halves. The calibrated tests are compatible with the fitted model at the stated level.

Table 15: Parameter and reliability estimates with 95 percent reliability intervals. Interval types are explained in the text.

Data	Estimator	α	a	b	R	95 percent interval
Jute	MLE	1.166	2.117	4.846	0.753	[0.444, 0.945]
Jute	Ordered MLE	1.166	2.117	4.846	0.753	[0.500, 0.947]
Jute	Bayes	1.179	2.201	3.901	0.666	[0.380, 0.895]
Jute	Ordered Bayes	1.165	2.104	4.127	0.699	[0.516, 0.899]
Carbon	MLE	2.874	1.341	0.695	0.316	[0.133, 0.615]
Carbon	Ordered MLE	2.660	1.135	1.135	0.500	[0.500, 0.862]
Carbon	Bayes	2.784	1.317	0.827	0.362	[0.159, 0.608]
Carbon	Ordered Bayes	2.552	1.080	1.338	0.565	[0.502, 0.704]

Table 16: Numerical 95 percent HPD intervals for reliability in the data examples. The probability column checks mass under the interpolated posterior density.

Data	Posterior	HPD interval	Width	Probability
Jute	Unrestricted	[0.4040, 0.9115]	0.5075	0.950
Jute	Ordered	[0.5000, 0.8744]	0.3744	0.950
Carbon	Unrestricted	[0.1462, 0.5915]	0.4454	0.950
Carbon	Ordered	[0.5000, 0.6752]	0.1752	0.950

Table 15 uses a log-ratio asymptotic interval for the unrestricted MLE, a 500-resample percentile bootstrap interval for the ordered MLE, and equal-tail intervals for the Bayesian rows. Table 16 gives the numerical HPD alternatives for the same posterior distributions. Confidence intervals and posterior credible intervals have different interpretations.

The jute estimates illustrate the calculation, but the rejected fit in one group must remain part of their interpretation. The carbon results are also sensitive to imposing an order: the ordered model excludes every reliability below one half. A narrower interval in that model is partly a consequence of the assumption, rather than evidence that the assumption is correct.

Table 17: Jute parameter estimates with positive asymptotic intervals for MLE and equal-tail intervals for the Bayesian rows. These intervals refer to the retained-data likelihood.

Parameter	Estimator	Estimate	95 percent interval
α	Bayes	1.1788	[0.7933, 1.6159]
α	Ordered Bayes	1.1654	[0.7933, 1.6159]
α	MLE	1.1656	[0.7901, 1.7195]
a	Bayes	2.2013	[1.0662, 3.9419]
a	Ordered Bayes	2.1039	[1.0347, 3.6596]
a	MLE	2.1167	[1.0223, 4.3826]
b	Bayes	3.9011	[1.5661, 7.2697]
b	Ordered Bayes	4.1271	[1.9756, 7.4257]
b	MLE	4.8465	[2.1311, 11.0215]

4.2.5 What the fitted estimates and posterior graphs show

For jute, the unrestricted MLE reliability is about 0.7526 and the unrestricted posterior mean is about 0.6656. The Bayesian estimate averages reliability over a posterior distribution rather than evaluating reliability at posterior mean rates. The difference between MLE and Bayes here reflects both the prior and the remaining uncertainty after censoring and record selection.

Table 18: Jute Bayesian reliability estimates under squared error and two LINEX loss choices. The posterior distribution is unchanged across loss choices in a row.

Posterior	Squared error	LINEX $c = 1.5$	LINEX $c = -1.5$
Unrestricted	0.6656	0.6516	0.6789
Ordered	0.6989	0.6906	0.7072

The ordered jute posterior mean is about 0.6989. This shift is partly imposed by excluding reliability below one half. In the carbon example the contrast is larger: the unrestricted and ordered Bayesian means are about 0.3623 and 0.5653. A change that large should be reported as sensitivity to an assumption, rather than as automatic improvement.

The intervals are wide because only a small part of the original sample enters the censored/record likelihood. In particular, six stress records contain less information than all thirty comparison measurements. The parameter posterior plots show skewness in the rates; this is one reason to retain full uncertainty rather than report only a point estimate.

The LINEX values in Table 18 use the same fitted posterior in each row but change the cost of overestimation relative to underestimation. They are alternative decisions under a loss function, not extra data fits. Squared error remains the main Bayesian comparison in the simulation.

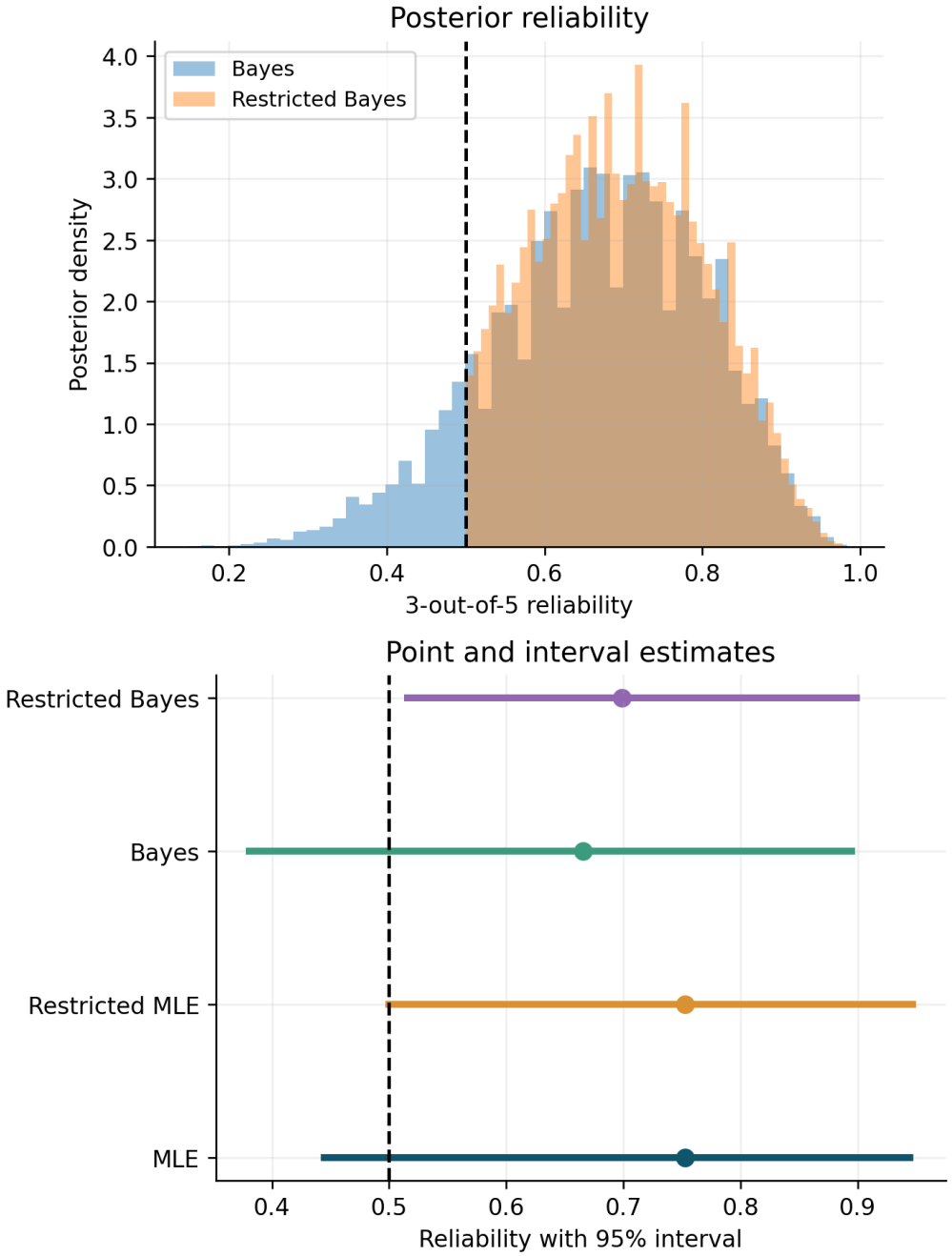


Figure 18: Jute posterior reliability histograms and the four point/interval estimates. Histogram draws come from the fitted posterior; interval types are identified in the text.

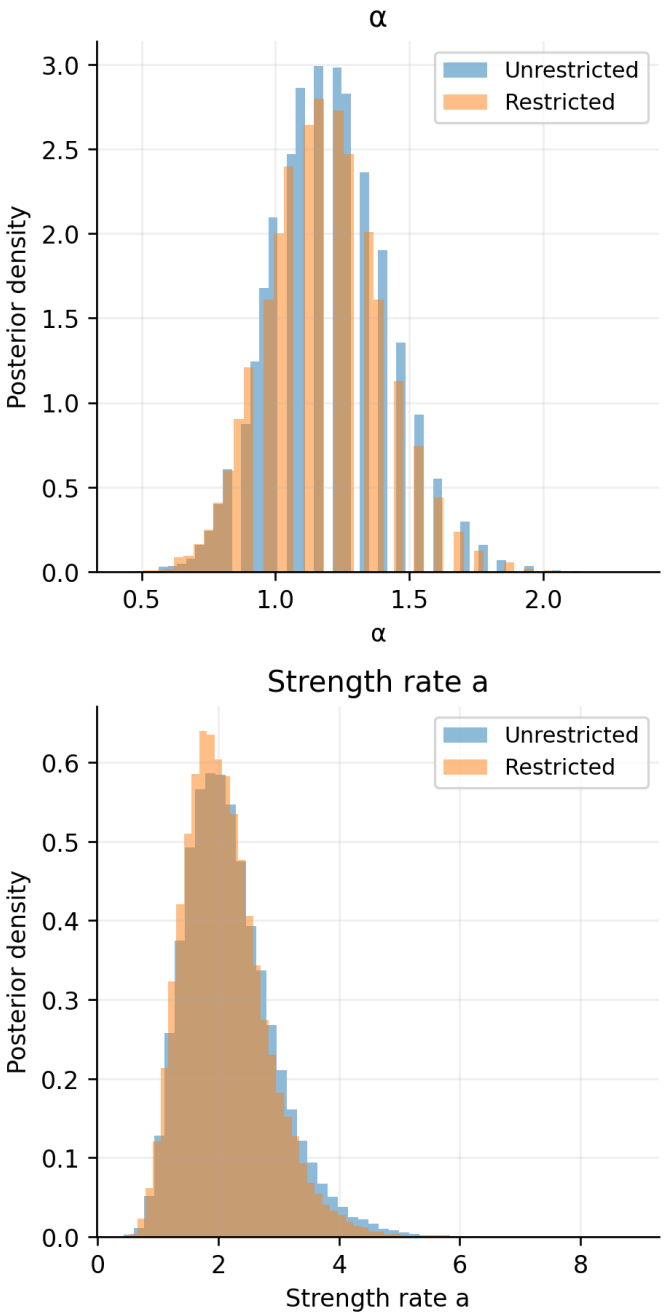


Figure 19: Jute posterior distributions of the common shape and strength rate, comparing unrestricted and ordered supports.

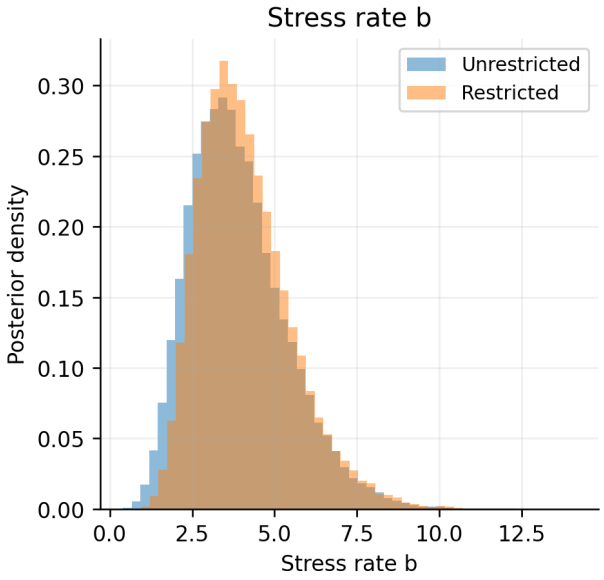


Figure 20: Jute stress-rate posterior distributions under the two supports. Rate values refer to the fixed measurement unit used in the analysis.

4.3 Checking the calculations

There are two different questions here: whether the numerical calculations are stable, and whether the Weibull model fits the measurements. Stable calculations do not make a poorly fitting model true. We therefore keep the numerical checks separate from the goodness-of-fit results.

We checked the reliability product against direct integration of the conditional binomial probability and against complete-system simulation. The implementation of the observed information was compared with a finite-difference Hessian. The ordered profile was also compared with a separate constrained optimization. These checks reduce the chance of a coding or algebra error in the main estimates.

Quadrature checks compare the selected numerical grid with a finer grid and examine the normalized mass near the shape integration limits. The data applications use a finer grid than the main simulation because there are only two examples to compute. Numerical HPD probability mass is checked against 0.95. Close agreement under grid refinement checks the numerical approximation used here; it is not a statement that all model assumptions are correct.

Table 19: Numerical comparisons used to check the calculation. Probability-mass checks refer to the interpolated density used for numerical HPD intervals.

Comparison	Largest difference	Scope
Reliability integral versus product	4.22×10^{-15}	Same theoretical target
Direct-system simulation versus product	0.00196	Within the stated Monte Carlo checks
Simulation reliability means, finer grid	1.37×10^{-3}	One dataset per scenario and support
Application means, finer grid	1.38×10^{-14}	Both posterior supports
Application equal-tail endpoints	4.65×10^{-5}	Both posterior supports
Application HPD endpoints	3.72×10^{-8}	Both posterior supports
HPD posterior probability versus 0.95	5.84×10^{-12}	All saved Bayesian simulation fits

The independent posterior sampler uses four chains per model, with 15,000 iterations each and the first 4,000 removed as warm-up. Proposals are made in transformed coordinates so that positive parameters and the order support are respected. Conditional draws of the total rate use its Gamma distribution. The trace figures display every tenth retained iteration to keep the lines readable. The stored chains retain the full draws.

Table 20: Checks from four retained chains for each model. Difference means sampler mean minus quadrature mean. R-hat is the classical split statistic.

Model	Target	R-hat	ESS	Mean	MCSE	Difference
Unrestricted	α	1.0006	5645	1.1722	0.0029	-0.0066
Unrestricted	a	1.0002	9428	2.1893	0.0076	-0.0120
Unrestricted	b	1.0009	8607	3.8709	0.0159	-0.0302
Unrestricted	R	1.0005	5996	0.6650	0.0017	-0.0006
Ordered	α	1.0001	6700	1.1635	0.0026	-0.0019
Ordered	a	1.0001	10088	2.0967	0.0067	-0.0072
Ordered	b	1.0002	7673	4.1293	0.0162	0.0021
Ordered	R	1.0007	3977	0.7003	0.0017	0.0014

A trace without a long visible drift is a useful check, but it is not proof of convergence. We also compare between-chain and within-chain variation through split R-hat, and report effective sample size and Monte Carlo error. Here the largest split R-hat is about 1.0009 and the smallest effective sample size is about 3977. The sampler means agree with quadrature within the stated Monte Carlo checks. These are classical split diagnostics rather than rank-normalized versions.

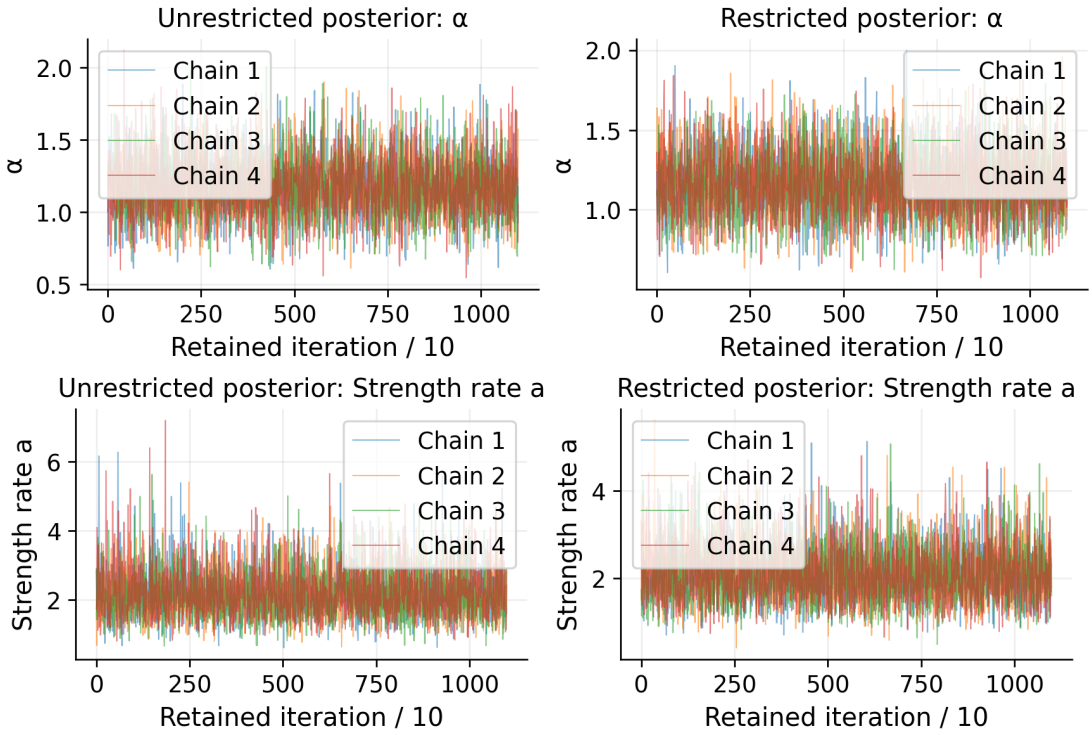


Figure 21: Four retained chains for shape (upper row) and strength rate (lower row). Unrestricted chains are on the left and ordered chains on the right. Every tenth retained state is plotted.

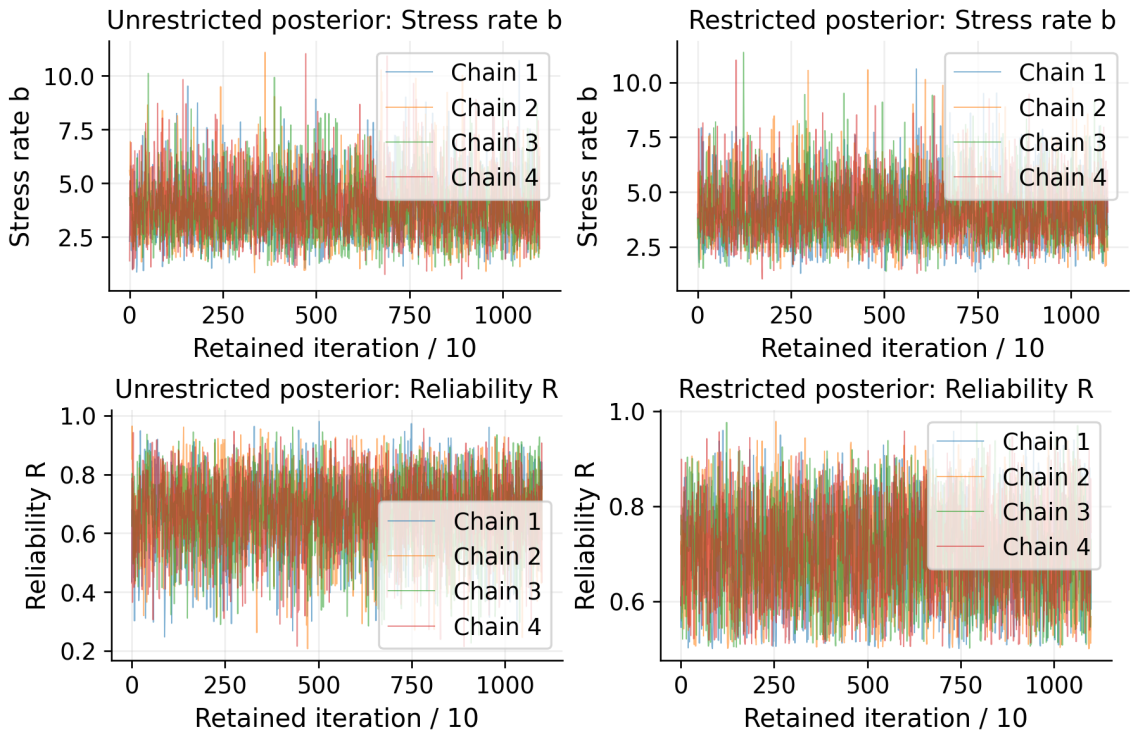


Figure 22: Four retained chains for stress rate (upper row) and reliability (lower row), using the same left/right support ordering.

4.4 Sensitivity and posterior prediction

A single fitted estimate does not show how much the answer depends on modelling choices. We vary one stated feature at a time in the jute example. Table 21 distinguishes changes to the prior, the hybrid stopping scheme and the record design. Ordered and unrestricted rows must still be read as different assumptions about the populations.

Table 21: Jute sensitivity calculations. All intervals in this table are posterior equal-tail intervals; each row changes the stated input.

Change	Setting	Ordered	R	95 percent interval
Prior	Primary	No	0.6656	[0.3805, 0.8947]
Prior	Primary	Yes	0.6989	[0.5157, 0.8988]
Prior	Broader total rate	No	0.6757	[0.3886, 0.9018]
Prior	Broader total rate	Yes	0.7063	[0.5173, 0.9053]
Prior	More concentrated rates	No	0.6194	[0.3639, 0.8474]
Prior	More concentrated rates	Yes	0.6609	[0.5103, 0.8546]
Hybrid scheme	Type I	No	0.6656	[0.3805, 0.8947]
Hybrid scheme	Type I	Yes	0.6989	[0.5157, 0.8988]
Hybrid scheme	Adaptive II	No	0.5450	[0.2924, 0.7863]
Hybrid scheme	Adaptive II	Yes	0.6223	[0.5057, 0.8047]
Record design	k=1, r=4	No	0.6782	[0.3564, 0.9183]
Record design	k=2, r=4	No	0.6454	[0.3321, 0.8949]
Record design	k=2, r=8	No	0.7116	[0.4479, 0.9126]
Record design	k=3, r=6	No	0.5856	[0.3110, 0.8393]

Under the three priors, the unrestricted posterior mean reliability ranges from about 0.6194 to 0.6757. The more concentrated rate prior pulls the estimate down compared with the primary prior. This is why the primary prior is described as a fixed proper prior, rather than as noninformative.

The result should not be presented as if it depended on the observations alone.

Changing the strength protocol from Type I to adaptive Type II changes the unrestricted posterior mean from about 0.6656 to 0.5450 in this retrospective example. The adaptive rule retains a different experiment; it is not just a different label on the same data. Similarly, different record orders and stopping counts select different stress information. Their estimates need not change monotonically. In particular, retaining more records changes both the information and the largest observed record.

The original printed order is only an imposed acquisition order. We therefore repeat record extraction after 200 random permutations of the complete comparison sample. Some permutations do not produce six records in the available sequence. The plotted distribution uses successful permutations only. Its spread shows acquisition-order sensitivity; it is not a confidence distribution for a new independent sample.

Of 200 attempted permutations, 123 reach the prescribed six upper 2-records and 77 do not. Among successful orders, posterior mean reliability ranges from 0.6025 to 0.8186, with mean 0.6587 and standard deviation 0.0356. The original printed-order mean is 0.6656.

For a posterior predictive check, we draw parameters from the fitted unrestricted posterior and generate a fresh experiment using the original censoring and record rules. We repeat this 2000 times and compare the retained failure count and largest stress record with the observed values. The orange vertical lines in the figures identify the observed values or the original acquisition-order estimate.

Table 22: Posterior predictive summaries from 2000 regenerated censored/record experiments. The last column is a descriptive upper-tail probability.

Statistic	Observed	95 percent predictive range	Upper tail
Observed failure count	13.0000	[8.0000, 19.0000]	0.666
Largest record	0.6627	[0.2974, 2.2573]	0.651

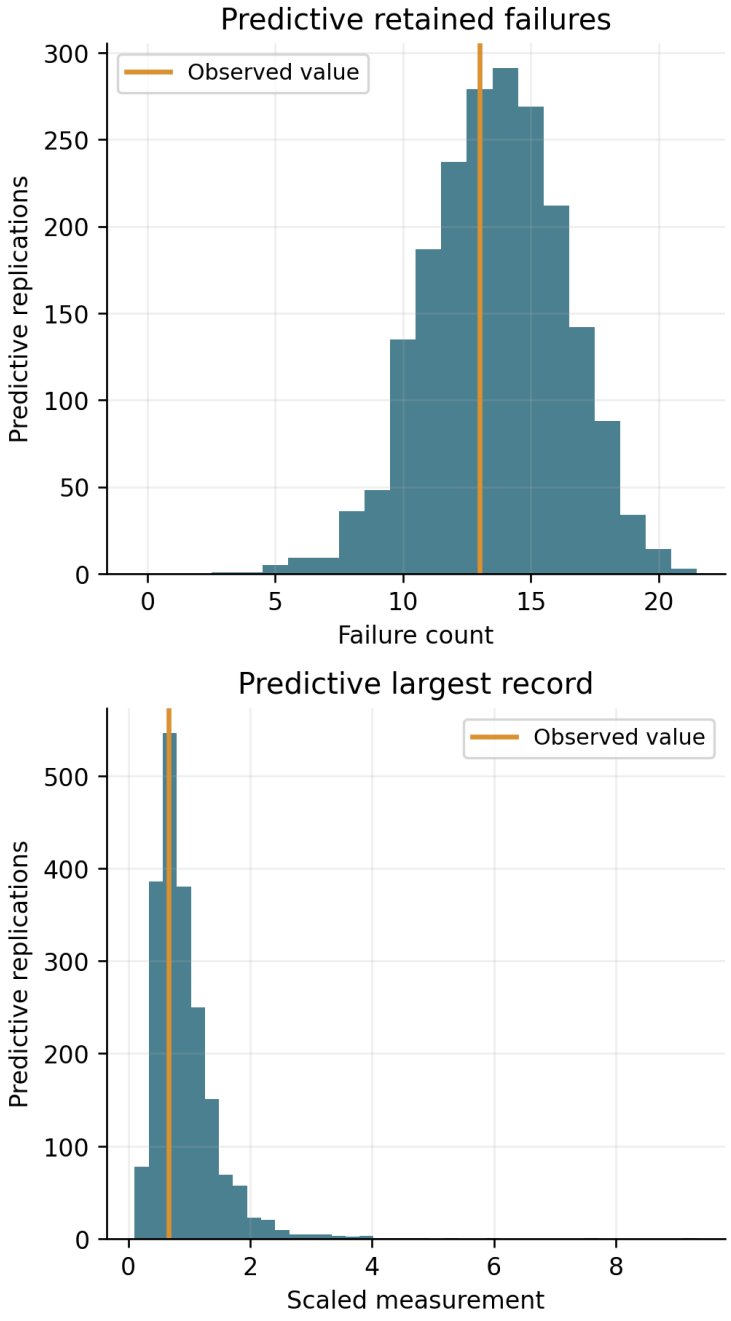


Figure 23: Posterior predictive distributions of retained failure count and largest stress record under the original jute observation protocol. Vertical lines mark the observed summaries.

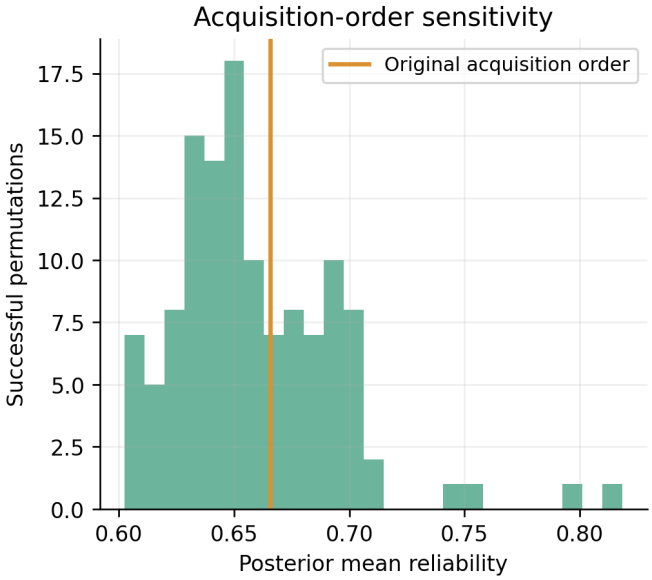


Figure 24: Posterior mean reliability after successful random acquisition-order permutations. The vertical line marks the original printed-order result.

The observed failure count, 13, falls within the predictive central range of 8 to 19. The largest retained stress record also falls within its predictive range. This says that these two summaries are not unusual under the fitted experiment. It does not cancel the complete-data jute goodness-of-fit rejection. Posterior predictive tail areas are descriptive model checks; they are not ordinary uniformly distributed significance-test p-values.

5 Conclusion

We derived the shared-stress reliability of a Weibull s-out-of-K system and used the appropriate likelihood for block progressive hybrid strengths and upper k-record stresses. Both unrestricted and ordered MLEs were obtained by profiling over the common shape. A dependent gamma-Dirichlet prior gave Bayesian estimates in both cases, with numerical integration reducing the posterior calculation to two dimensions.

The simulation compares point-estimation errors and the coverage and width of asymptotic, equal-tail and numerical HPD intervals. The ordered Bayesian estimates perform well in the correctly ordered cases studied here. A false restriction produces substantial bias, and intervals constrained above one half cannot cover a reliability below one half. At the equality boundary, HPD and equal-tail intervals behave differently because an HPD interval may include the boundary itself.

Published measurements provide reproducible illustrations, rather than proof of engineering performance. The jute example has a documented fit problem; the carbon example gives a compatible additional validation. The common-shape assumption, noninformative withdrawals and the imposed record acquisition order are limitations. The notebook includes prior and acquisition-order sensitivity checks and posterior diagnostics. Future work could study unequal shapes or other withdrawal plans, but those extensions are outside the present project.

A Complete simulation settings

Table 23 lists every scenario. Ordered, boundary and violated rate pairs are $(0.8, 1.6)$, $(1, 1)$ and $(1.6, 0.8)$, respectively. In every row the target is $\lfloor 0.7n \rfloor$ per block, the withdrawal plan is balanced and the threshold is the true strength 65th percentile. The record count increases with block size. All point and interval summaries for individual scenarios are stored in `data/results/simulation_summary.csv`; the raw fitted replications are in `data/results/simulation_raw.csv`.

Table 23: Complete 48-scenario design. Every row has three strength blocks and 300 independent replications. O, B and V mean ordered, boundary and violated.

ID	α	n	Protocol	Regime	k	r
1	0.8	20	Type I	O	1	6
2	0.8	20	Type I	O	3	6
3	0.8	20	Adaptive II	O	1	6
4	0.8	20	Adaptive II	O	3	6
5	0.8	40	Type I	O	1	12
6	0.8	40	Type I	O	3	12
7	0.8	40	Adaptive II	O	1	12
8	0.8	40	Adaptive II	O	3	12
9	1.8	20	Type I	O	1	6
10	1.8	20	Type I	O	3	6
11	1.8	20	Adaptive II	O	1	6
12	1.8	20	Adaptive II	O	3	6
13	1.8	40	Type I	O	1	12
14	1.8	40	Type I	O	3	12
15	1.8	40	Adaptive II	O	1	12
16	1.8	40	Adaptive II	O	3	12
17	0.8	20	Type I	B	1	6
18	0.8	20	Type I	B	3	6
19	0.8	20	Adaptive II	B	1	6
20	0.8	20	Adaptive II	B	3	6
21	0.8	40	Type I	B	1	12
22	0.8	40	Type I	B	3	12
23	0.8	40	Adaptive II	B	1	12
24	0.8	40	Adaptive II	B	3	12
25	1.8	20	Type I	B	1	6
26	1.8	20	Type I	B	3	6
27	1.8	20	Adaptive II	B	1	6
28	1.8	20	Adaptive II	B	3	6
29	1.8	40	Type I	B	1	12
30	1.8	40	Type I	B	3	12
31	1.8	40	Adaptive II	B	1	12
32	1.8	40	Adaptive II	B	3	12
33	0.8	20	Type I	V	1	6
34	0.8	20	Type I	V	3	6
35	0.8	20	Adaptive II	V	1	6
36	0.8	20	Adaptive II	V	3	6
37	0.8	40	Type I	V	1	12
38	0.8	40	Type I	V	3	12
39	0.8	40	Adaptive II	V	1	12

ID	α	n	Protocol	Regime	k	r
40	0.8	40	Adaptive II	V	3	12
41	1.8	20	Type I	V	1	6
42	1.8	20	Type I	V	3	6
43	1.8	20	Adaptive II	V	1	6
44	1.8	20	Adaptive II	V	3	6
45	1.8	40	Type I	V	1	12
46	1.8	40	Type I	V	3	12
47	1.8	40	Adaptive II	V	1	12
48	1.8	40	Adaptive II	V	3	12

B Complete published observations

These tables reproduce the measurements used in the two applications so that the data selection can be checked without reading Python. Their source and scaling are described in Section 4.2. No observed measurement in these tables is a newly simulated value.

Table 24: All 30 measurements in each jute group, in the source table order and with the source scaling already applied.

Position	15 mm	20 mm
1	0.59440	0.07146
2	0.20275	0.41902
3	0.16837	0.28464
4	0.57486	0.58557
5	0.22565	0.45660
6	0.07638	0.11385
7	0.15667	0.18785
8	0.12781	0.68816
9	0.81387	0.66266
10	0.56239	0.04558
11	0.46847	0.57862
12	0.13509	0.75670
13	0.07224	0.59429
14	0.49794	0.16649
15	0.35556	0.09972
16	0.56907	0.70736
17	0.64048	0.76514
18	0.20076	0.18713
19	0.55042	0.14596
20	0.74875	0.35070
21	0.48966	0.54744
22	0.67806	0.11699
23	0.45771	0.37581
24	0.10673	0.58160
25	0.71630	0.11986
26	0.04266	0.04801
27	0.08040	0.20016
28	0.33922	0.03675
29	0.07009	0.24453
30	0.19342	0.08355

Table 25: All 100 carbon-fibre observations in original printed order, in GPa. Each row contains consecutive source positions.

Positions	1	2	3	4	5	6	7	8
1–8	3.70	2.74	2.73	2.50	3.60	3.11	3.27	2.87
9–16	1.47	3.11	4.42	2.41	3.19	3.22	1.69	3.28
17–24	3.09	1.87	3.15	4.90	3.75	2.43	2.95	2.97
25–32	3.39	2.96	2.53	2.67	2.93	3.22	3.39	2.81
33–40	4.20	3.33	2.55	3.31	3.31	2.85	2.56	3.56
41–48	3.15	2.35	2.55	2.59	2.38	2.81	2.77	2.17
49–56	2.83	1.92	1.41	3.68	2.97	1.36	0.98	2.76
57–64	4.91	3.68	1.84	1.59	3.19	1.57	0.81	5.56
65–72	1.73	1.59	2.00	1.22	1.12	1.71	2.17	1.17
73–80	5.08	2.48	1.18	3.51	2.17	1.69	1.25	4.38
81–88	1.84	0.39	3.68	2.48	0.85	1.61	2.79	4.70
89–96	2.03	1.80	1.57	1.08	2.03	1.61	2.12	1.89
97–100	2.88	2.82	2.05	3.65				

C Source use and reproducibility

The Weibull block paper [1] supplies the relevant common-shape parameterization, order-restricted inference and gamma-Dirichlet prior construction. Its reproduced jute table is our transcription source. The Scientific Reports paper [2] supplies the combined censoring/record framework for a different distribution. We derive the Weibull reliability and likelihood separately. The Journal of Mathematics paper [3] gives related conditional reliability and bootstrap ideas and supplies the reproduced carbon measurements.

Two other supplied articles were reviewed. The generalized inverted exponential paper by Hasaballah and coauthors concerns unified hybrid inference and provides background on interval reporting; its distribution and COVID-19 data are not used here. The attention-based ratio control-chart article by Yeganeh and coauthors is outside the assigned reliability problem, so none of its trading data or machine-learning procedures is imported. The assigned SAGE reference was available only at abstract level. Its full text was not supplied, and no derivation from its unavailable full text is claimed. The detailed source-use notes remain in `code/notes/literature_review.md`.

The repository separates raw measurements, retained observations and computed results. The notebook in `code/` contains saved tables and graphs. `code/analysis.py` contains the statistical functions; the other scripts execute the notebook and prepare the report tables and graphs. The complete simulation can be repeated by enabling its stated run switch. The master seed and scenario-specific streams are fixed. The LaTeX sources and bibliography are in `latex/`, and every graph is in `graphs/`.

The main results do not depend on reading the locally archived original article PDFs at runtime. Those documents remain available locally for source checking, but are not part of the public repository. This separation keeps the submitted project focused on the report, code, data and computed evidence.

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